

A Universal Geometric Theory of Nuclear Reactions in CPTG

PRIMAT-Anchored Reduced-Graph Reachability and the Complete $A = 1$ –119 Mass-Sector Register

Carter L Glass Jr

Abstract

This computational companion tests whether the fixed geometric nuclear-reaction framework developed in CPTG can be translated into a reproducible network calculation that extends beyond the native primordial domain without inserting a heavy-sector seed or tuning the continuation to a desired endpoint. PRIMAT v0.3.2 supplies the native trajectory and endpoint authority through $A = 23$. From that boundary, a fixed native-to-external source rule drives a selected ReacLib forward capture and beta-minus graph. The principal result is a complete mass-sector register from $A = 1$ through $A = 119$: $A = 5$ is retained as a physical-gap annotation, while every external sector from $A = 24$ through 119 is structurally reachable and has strictly positive modeled support at all six PRIMAT baryon-density anchors.

The calculation is subjected to independent numerical qualification rather than being accepted from a single transport scheme. A separate prospective PRIMAT-native implementation also tests the locked CPTG mass-seven transport law directly in the live network, reproducing the fixed $1/\pi$ survival with a worst relative deviation of 0.018083% while leaving the principal light-element observables essentially unchanged. Beyond the principal $A = 1$ –119 register, the same reduced-graph rule identifies a natural reachable frontier at $A = 338$, with the eligible $A = 339$ sector disconnected under the selected topology. Source-tail truncation shows no resolved relative change larger than 2.36×10^{-13} in the unrenormalized $A = 338$ endpoint when the faint late boundary-source tail is removed.

The value of this companion is methodological as well as numerical: it converts the CPTG nuclear-reaction construction into an explicit computational object that can be inspected, rerun, falsified, and compared against a native network authority while preserving a strict separation between structural reachability and abundance claims. The external Y_A values are prescribed-bath reduced-transport populations, not self-consistent final abundances of a coupled PRIMAT heavy-element network and not asserted primordial heavy-element yield predictions.

1 Introduction

This paper is the computational companion to *A Universal Geometric Theory of Nuclear Reactions in CPTG: The Four-Sector Foundation of Vertex, Bridge, Closure, and Saturation* [1]. The parent paper develops the geometric structure of the theory; the commissioned deuterium–proton capture and primordial mass-seven study [2] establishes the reaction-specific foundation from which the broader nuclear program emerged. The role of this companion is different. It asks what the theory becomes when it is required to operate as a concrete, auditable nuclear-network calculation.

That distinction is important. A geometric reaction law has scientific value only if its consequences can be translated into explicit network coordinates, propagated without hidden endpoint fitting, and tested against an independent numerical authority. The purpose of this manuscript is therefore to build that

translation layer. It fixes a native primordial trajectory authority, defines the boundary at which the native calculation ends, constructs the permitted source into an external reaction graph, propagates that source with independently checked numerical methods, and records the resulting mass-sector structure in a form that can be reproduced and falsified. The calculation is designed to expose where the geometric construction succeeds, where the selected reaction topology ends, and which conclusions belong to native network physics versus reduced-graph post-processing.

The central deliverable is the complete $A = 1$ –119 mass-sector register. It joins the native light and intermediate PRIMAT inventory to a seed-free external continuation and asks a deliberately simple but demanding question: can every integer mass sector in the declared range be reached and supported by the fixed construction without importing a heavy-sector endpoint vector or inserting an artificial abundance floor? The companion also performs two complementary tests that illuminate the meaning of that register. First, the locked CPTG mass-seven transport law is implemented prospectively inside PRIMAT itself, providing a direct native-network test of a CPTG transport prediction rather than a post-processing adjustment. Second, the same reduced-graph rule is extended beyond $A = 119$ to determine its natural structural frontier and to test whether the extreme frontier response depends on a faint late-time source tail.

These tasks give the companion a broader function than tabulating calculated populations. It provides a controlled interface between geometric theory and reaction-network dynamics. Structural reachability, numerical convergence, source robustness, and physical abundance authority are kept as separate questions rather than being collapsed into a single claim. This separation is particularly important in the post-silicon region, where a positive reduced-graph population can establish that a pathway exists within the declared operator without implying that a complete heavy-element nucleosynthesis network has been solved. The resulting framework therefore supplies both positive evidence and explicit failure conditions: the graph can terminate, a numerical sequence can fail qualification, a source test can fail, or a native-network implementation can disagree with the locked prediction.

The primary native authority used here is PRIMAT v0.3.2 [3]. Six baryon-density trajectories are fixed at

$$\eta_{10} \in \{6.044, 6.080, 6.094, 6.106, 6.128, 6.134\}. \quad (1)$$

In the exact large-network configuration used for this work, the native species set contains 59 nuclei and reaches a maximum mass number $A = 23$. This is a property of the declared campaign configuration, not a statement about every possible PRIMAT network. The external construction therefore begins at $A = 24$, using only the frozen native histories and a rate-library graph generated under the rules defined below. No heavy-sector endpoint vector from another transport calculation is a scientific input.

The complete register contains one entry for every integer mass number

$$1 \leq A \leq 119. \quad (2)$$

The only non-abundance entry in the native interval is $A = 5$. PRIMAT contains no bound $A = 5$ abundance coordinate in the frozen species set; the unbound ${}^5\text{He}$ and ${}^5\text{Li}$ resonances are therefore retained as physical-gap annotations [6]. Every external mass sector from $A = 24$ through 119 contains at least one reachable graph coordinate and has a strictly positive modeled mass-sector sum at all six anchors.

Numerical qualification is treated as part of the scientific claim rather than as an implementation detail. The transport discretization is tested with independent primary and backward-Euler schemes and preregistered refinement sequences before the production result is interpreted. The production calculation separately checks finite nonnegative states, exact-zero-source behavior, source linearity,

reconstruction consistency, and complete mass-sector support. Detailed gates and checkpoint results are given in the numerical sections below; the evidence objects, execution identities, and SHA-256 records are consolidated in Section A.

The scope of the paper is consequently precise. PRIMAT supplies native trajectory and endpoint authority through $A = 23$. The $A = 24\text{--}119$ result is a prescribed-PRIMAT-bath reduced-graph transport calculation. The post- $A = 119$ calculations test the same reduced operator for continuum behavior, structural reachability, and source-tail robustness; they do not convert it into native PRIMAT heavy-network evolution. The claim boundary is

For this frozen campaign, PRIMAT supplies the native $A \leq 23$ trajectory and endpoint authority. The $A = 24\text{--}119$ result is a prescribed-PRIMAT-bath reduced-graph transport calculation, not native PRIMAT heavy-network evolution and not a self-consistent coupled primordial heavy-element yield calculation.

2 Relation to the CPTG Framework

The detailed CPTG source-current, curvature, incidence, and reaction-source formulas are developed in the parent universal-theory paper and are not repeated here. The present computational companion does not refit those formulas. Its task is to document and test a downstream numerical operator once the native PRIMAT trajectory has been fixed.

For a native or external nuclear coordinate i , Y_i denotes the number-abundance coordinate used by the reaction-network calculation. The mass-sector population is

$$Y_A = \sum_{i:A_i=A} Y_i. \quad (3)$$

For native PRIMAT rows, Equation (3) is a sum of the frozen C-backend final abundances. For external rows it is a sum of the post-processed transport state. Those two uses share a numerical coordinate but not the same authority: the external state is driven by prescribed native histories and is not fed back into the PRIMAT network.

The computational architecture is

$$\begin{array}{c}
 \text{frozen PRIMAT native solution} \\
 \downarrow \\
 \text{native-to-external boundary source} \\
 \downarrow \\
 \text{selected forward reduced graph} \\
 \downarrow \\
 \text{positive transport state and mass-sector register.}
 \end{array} \quad (4)$$

The external step is one-way post-processing. The native heavy source coordinates and the light-particle bath are not depleted by the external transport. Consequently, a combined sum of native and external

entries must not be interpreted as the output of one closed, baryon-conserving coupled network, even though every retained reaction edge obeys its nuclear A - and Z -bookkeeping rule.

3 Mass Seven as a Native Transport Sector

Mass seven occupies a distinctive position in the present calculation. It lies entirely inside the native PRIMAT domain of the frozen campaign, and it is also the first abundance-transport sector for which the locked CPTG transport law is implemented prospectively inside the primary PRIMAT authority. Two numerical objects therefore remain deliberately distinct: the *ungated native PRIMAT mass-seven coordinate* used by the frozen complete-chain register and the *separate gated PRIMAT live-channel validation* used to test the CPTG mass-seven law. The gated validation is not used to modify the native trajectories, the $A = 1\text{--}119$ register, or the boundary sources used by the external continuation.

For the frozen complete-chain campaign, the $A = 7$ sector contains the two native abundance coordinates

$${}^7\text{Li}, \quad {}^7\text{Be}, \quad (5)$$

and its mass-sector population is

$$Y_7 = Y_{{}^7\text{Li}} + Y_{{}^7\text{Be}}. \quad (6)$$

At the central complete-chain anchor,

$$\eta_{10} = 6.094, \quad (7)$$

the frozen C-backend register gives

$$Y_7 = 4.080752 \times 10^{-10}, \quad (8)$$

with the six-anchor envelope

$$4.01174 \times 10^{-10} \leq Y_7 \leq 4.13619 \times 10^{-10}. \quad (9)$$

Thus $A = 7$ is an ordinary abundance-bearing native mass sector in the complete-chain register. Unlike $A = 5$, it is not a resonance-only annotation, and unlike the $A \geq 24$ continuation, it does not depend on the external prescribed-bath transport operator.

The use of a mass-sector coordinate is especially natural at $A = 7$. During primordial nucleosynthesis, a substantial part of the surviving mass-seven inventory is carried by ${}^7\text{Be}$ and later contributes to lithium after electron capture. The relevant network coordinate is therefore the surviving sum in Equation (6), rather than an isolated instantaneous ${}^7\text{Li}$ coordinate. This provides a direct physical example of why the present work organizes the complete calculation by mass sector while retaining individual nuclides inside each sector.

3.1 Locked CPTG mass-seven law

The CPTG mass-seven construction fixes the live transport optical depth

$$\tau_7 = \int \Gamma_{7,\text{trans}}(t) dt = \ln \pi, \quad (10)$$

with direct live-channel evolution

$$\frac{dY_{7\text{Li}}}{dt} \longrightarrow \frac{dY_{7\text{Li}}}{dt} - \Gamma_{7,\text{trans}}(t)Y_{7\text{Li}}, \quad (11)$$

$$\frac{dY_{7\text{Be}}}{dt} \longrightarrow \frac{dY_{7\text{Be}}}{dt} - \Gamma_{7,\text{trans}}(t)Y_{7\text{Be}}. \quad (12)$$

For an isolated survival channel the fixed optical depth implies

$$Y_{7,\text{CPTG}} = Y_{7,\text{raw}}e^{-\tau_7} = \frac{Y_{7,\text{raw}}}{\pi}, \quad (13)$$

with

$$f_7 = \frac{1}{\pi} = 0.3183098861837907. \quad (14)$$

The law is a live source-network transport term, not a stellar depletion coefficient and not a post-processing division of a completed abundance. The earlier commissioned study retained this distinction between the reaction-specific deuterium–proton construction and the independent mass-seven transport law [2].

3.2 Prospective PRIMAT-native implementation

The prospective validation treats Equations (10)–(12) as locked theory. No lithium endpoint was used to set the survival factor, gate normalization, gate window, baryon coordinate, nuclear rates, neutron lifetime, or solver tolerance. A copied PRIMAT v0.3.2 source tree was used so that the installed authority tree remained byte-for-byte unchanged. The run forced the Python backend, used the native large network, numerical precision 10^{-10} , a 4000-point nuclear-rate grid, neutron lifetime 878.4 s, and the same standard radiative, finite-mass, thermal, spectral-distortion, weak-rate-normalization, plasma-QED, and nuclear-QED corrections adopted by the native campaign.

The direct sink species were exactly ${}^7\text{Li}$ and ${}^7\text{Be}$. Matching diagonal sink terms were inserted into PRIMAT’s analytic Jacobian. The temporal embedding was prospectively frozen as a log-cosine window from 0.0175 to 0.0040 MeV, normalized to the optical depth in Equation (10). Seven paired coordinates were then executed gate-off and gate-on, giving fourteen native solves: one CPTG transported baryon-density coordinate with

$$\Omega_b h^2 = 0.021898765370 \quad (15)$$

and the six frozen authority anchors in Equation (1). Every row used $\Delta N_{\text{eff}} = 0$. The separate CPTG radiation-coordinate mapping was not implemented in this campaign and is not inferred from that setting.

The hard acceptance gates were frozen before gated endpoint exposure. They required the internal Y_7 and reported mass-seven/H survival ratios to agree with $1/\pi$ within 0.5% relative; gated-vs-baseline changes in D/H, Y_p , and ${}^3\text{He}/H$ no larger than 10^{-6} ; an independent-grid optical-depth reconstruction within 2×10^{-6} relative of $\ln \pi$; nonzero runtime exercise of both the RHS and analytic-Jacobian sink paths; finite/nonnegative native final states; unchanged installed PRIMAT source; and successful reproduction of all six ungated PRIMAT authority baselines before gate-on qualification. The observational lithium comparison was recorded only as a downstream diagnostic and could not alter the scientific decision.

3.3 Native live-channel result

All seven paired coordinates passed. The measured PRIMAT live-network survival factors are listed in Table 1.

Coordinate	Y_7 survival	rel. error to $1/\pi$	D/H rel. change	$^3\text{He}/H$ rel. change
CPTG BBN baryon	0.318367445312	1.80827×10^{-4}	3.14×10^{-9}	2.02×10^{-10}
$\eta_{10} = 6.044$	0.318366185395	1.76869×10^{-4}	3.63×10^{-9}	5.89×10^{-10}
$\eta_{10} = 6.080$	0.318365290272	1.74057×10^{-4}	3.16×10^{-9}	2.88×10^{-10}
$\eta_{10} = 6.094$	0.318364947655	1.72981×10^{-4}	3.87×10^{-9}	5.69×10^{-10}
$\eta_{10} = 6.106$	0.318364656240	1.72065×10^{-4}	3.23×10^{-9}	3.42×10^{-10}
$\eta_{10} = 6.128$	0.318364128306	1.70407×10^{-4}	3.29×10^{-9}	2.15×10^{-10}
$\eta_{10} = 6.134$	0.318363988002	1.69966×10^{-4}	2.79×10^{-9}	2.16×10^{-11}

Table 1: Prospective PRIMAT-native mass-seven live-channel validation. All rows pass the frozen theory and non- $A = 7$ control gates. The observational lithium comparison is not an acceptance criterion.

The complete survival interval is

$$0.318363988002 \leq \frac{Y_{7,\text{gated}}}{Y_{7,\text{baseline}}} \leq 0.318367445312, \quad (16)$$

and the worst relative deviation from the locked target is

$$\max \left| \frac{S_7 - (1/\pi)}{1/\pi} \right| = 1.80827 \times 10^{-4} = 0.018083\%. \quad (17)$$

The largest non-mass-seven disturbances over all seven coordinates are

$$\max |\Delta(\text{D}/H)|_{\text{rel}} = 3.87 \times 10^{-9}, \quad (18)$$

$$\max |\Delta Y_p|_{\text{rel}} = 3.83 \times 10^{-12}, \quad (19)$$

$$\max |\Delta(^3\text{He}/H)|_{\text{rel}} = 5.89 \times 10^{-10}. \quad (20)$$

The independent 32,769-node optical-depth reconstruction has relative error 1.5×10^{-7} to $\ln \pi$ at every coordinate. Runtime instrumentation records 15,314–15,581 nonzero RHS gate calls and 435–473 nonzero analytic-Jacobian gate calls per gated row, directly establishing that the transport terms were exercised during the native integration rather than merely present in source code.

At the central PRIMAT authority anchor, the forced-Python live-validation baseline and gated results are

$$Y_{7,\text{baseline}} = 4.08074346577861223 \times 10^{-10}, \quad (21)$$

$$Y_{7,\text{gated}} = 1.29916567987528558 \times 10^{-10}, \quad (22)$$

$$(^7\text{Li}/H)_{\text{baseline}} = 5.41925964547088552 \times 10^{-10}, \quad (23)$$

$$(^7\text{Li}/H)_{\text{gated}} = 1.72530231335814528 \times 10^{-10}. \quad (24)$$

The slight difference between the Python-backend baseline above and the C-backend register value

in Equation (8) is retained rather than conflated: the former belongs to the paired live-validation backend, while the latter remains the frozen endpoint coordinate used by the complete-chain register.

At the CPTG transported baryon-density coordinate, the paired native calculation gives

$$Y_{7,\text{baseline}} = 3.94409054504861126 \times 10^{-10}, \quad (25)$$

$$Y_{7,\text{gated}} = 1.25567003090824555 \times 10^{-10}, \quad (26)$$

$$({}^7\text{Li}/H)_{\text{baseline}} = 5.23668866791401501 \times 10^{-10}, \quad (27)$$

$$({}^7\text{Li}/H)_{\text{gated}} = 1.66719119310060441 \times 10^{-10}. \quad (28)$$

The downstream observational pull diagnostic spans 0.868765σ to 1.19553σ across the seven gated coordinates. Because observational lithium was excluded from the hard acceptance criteria, this agreement is a consequence of the locked transport law under the tested backgrounds rather than the quantity used to set or select that law.

Seven SciPy adaptive-quadrature warnings occur, one per gated coordinate. They report roundoff-limited certification at the requested 10^{-12} quadrature tolerance and do not abort any native solve. The separately preregistered independent-grid optical-depth reconstruction passes at 1.5×10^{-7} relative for every coordinate, so the warnings are recorded as numerical-certification warnings rather than failed gate normalizations.

The scientific decision is therefore

The locked CPTG $A = 7$ live-channel transport law passes prospective native implementation in PRIMAT v0.3.2's large network at the CPTG transported baryon-density coordinate and all six frozen PRIMAT authority anchors.

This does not constitute a fitted lithium correction, a post-processing division by π , a validation over arbitrary untested BBN backgrounds, or a full CPTG radiation-coordinate implementation in PRIMAT. The separate CPTG radiation-coordinate mapping is not implemented in this validation.

Within this companion, mass seven provides the following bridge across the mass-sector construction:

$$\begin{aligned} A = 1-4 &: \text{light-state foundation,} \\ A = 5 &: \text{physical resonance gap,} \\ A = 6 &: \text{return to bound abundance coordinates,} \\ A = 7 &: \text{PRIMAT-native live CPTG transport validated,} \\ A = 8-23 &: \text{continued native PRIMAT inventory,} \\ A \geq 24 &: \text{PRIMAT-anchored external reduced-graph continuation.} \end{aligned} \quad (29)$$

The ungated native trajectory is the controlling source for the complete-chain calculation, while the prospective live-channel validation establishes primary-network support for the locked mass-seven transport law without using that result to tune or modify the downstream heavy-sector construction.

4 PRIMAT Native Campaign Authority and Boundary Construction

4.1 Frozen native solution and backend roles

The controlling native trajectory authority contains six trajectory/meta pairs, one for each anchor in Equation (1). The exact configuration uses PRIMAT’s large network, numerical precision 10^{-10} , a 4000-point nuclear-rate grid over $10^{-3} \leq T_9 \leq 10$, neutron lifetime 878.4 s, and the radiative, finite-mass, thermal, spectral-distortion, neutron-lifetime-normalization, and nuclear-QED corrections enabled in the frozen configuration.

Two native backends have different roles in this companion. The C backend supplies the final native abundance dictionary used for the $A \leq 23$ register. The Python backend supplies the dense low-temperature solution required to construct time-dependent external sources. For each anchor, 513 logarithmically spaced samples are taken from approximately 94.843 s to 1.31565×10^6 s. At the central anchor this corresponds to approximately 1.2765×10^9 K down to 1.16045×10^7 K, with baryon mass density decreasing from approximately 5.5440×10^{-5} to 3.2081×10^{-11} g cm $^{-3}$.

The dense Python trajectory is accepted only if it is finite and has no component below -10^{-25} ; accepted values are then clipped at zero before external-source construction. The standard PRIMAT observables from the C backend reproduce the frozen campaign references exactly at stored precision, while the largest Python-versus-C standard-observable symmetric relative difference over the six anchors is 2.631735×10^{-6} , inside the preregistered 5×10^{-6} gate. The frozen species identity is the same at all six anchors: 59 native nuclear species with maximum $A = 23$.

4.2 Boundary-source rule and the fourteen realized crossings

The preregistered light-bath candidate set is

$$\mathcal{L} = \{n, p, {}^2\text{H}, {}^3\text{H}, {}^3\text{He}, {}^4\text{He}\}. \quad (30)$$

An allowed capture has one heavy reactant h with $A_h \geq 6$, exactly one light reactant $\ell \in \mathcal{L}$, and one nuclear product d satisfying

$$A_d = A_h + A_\ell, \quad Z_d = Z_h + Z_\ell. \quad (31)$$

A boundary-source edge additionally requires h to belong to the frozen native PRIMAT species set and d to lie outside that set with $A_d \geq 24$.

Applying this rule to a `pynucastro` 2.11.0 ReacLib library [4, 5] produces exactly fourteen native-to-external source edges, listed in Table 2. Although Equation (30) makes six light species available to the rule, the realized boundary crossings use only n , p , and ${}^4\text{He}$.

Boundary capture	External product
$^{20}\text{O}(^4\text{He}, \gamma)$	^{24}Ne
$^{20}\text{F}(^4\text{He}, \gamma)$	^{24}Na
$^{20}\text{Ne}(^4\text{He}, \gamma)$	^{24}Mg
$^{20}\text{Na}(^4\text{He}, \gamma)$	^{24}Al
$^{21}\text{Ne}(^4\text{He}, \gamma)$	^{25}Mg
$^{21}\text{Na}(^4\text{He}, \gamma)$	^{25}Al
$^{22}\text{Ne}(^4\text{He}, \gamma)$	^{26}Mg
$^{22}\text{Na}(^4\text{He}, \gamma)$	^{26}Al
$^{23}\text{Ne}(n, \gamma)$	^{24}Ne
$^{23}\text{Ne}(p, \gamma)$	^{24}Na
$^{23}\text{Ne}(^4\text{He}, \gamma)$	^{27}Mg
$^{23}\text{Na}(n, \gamma)$	^{24}Na
$^{23}\text{Na}(p, \gamma)$	^{24}Mg
$^{23}\text{Na}(^4\text{He}, \gamma)$	^{27}Al

Table 2: The fourteen realized native-to-external source edges. They are selected by the frozen topology rule; no external abundance seed is inserted.

The implementation enforces a nonnegative forward rate

$$r_e^+(T) = \max(r_e(T), 0). \quad (32)$$

For a boundary edge $h + \ell \rightarrow d + \gamma$, the external source is

$$S_d(t) += \rho_b(t)Y_\ell(t)r_e^+[T(t)]Y_h(t). \quad (33)$$

Every external state is initialized to exact zero. Equation (33) is an imposed post-processing source term: it does not subtract the corresponding reaction flow from the frozen PRIMAT heavy-state trajectory.

5 External Reaction Graph

The rate-library selection is constructed from `pynucastro` 2.11.0 for this calculation. Before reachability restriction it identifies 17,565 non-weak two-body one-product capture rates satisfying Equation (31) and 5,343 one-body beta-minus rates satisfying

$$A_d = A_s, \quad Z_d = Z_s + 1. \quad (34)$$

Starting from the fourteen boundary edges and restricting species to $A \leq 160$, the selected graph contains 2,927 external nuclear coordinates and 10,314 registered rows. Their accounting is

$$10,314 = 14 \text{ boundary-source} + 10,134 \text{ internal} + 166 \text{ outer-sink rows}. \quad (35)$$

By process type, 8,622 rows are captures and 1,692 are beta-minus edges. In the actual reachable graph all 8,622 capture rows use only neutron, proton, or ^4He reactants: 2,889 neutron captures, 2,851 proton captures, and 2,882 ^4He captures. No deuterium-, tritium-, or ^3He -capture row survives the selected topology and reachability construction.

For an internal edge e , define the nonnegative coefficient

$$\lambda_e(t) = \begin{cases} \rho_b(t)Y_\ell(t)r_e^+[T(t)], & e \text{ is a retained light-particle capture,} \\ r_e^+[T(t)], & e \text{ is a retained beta-minus decay.} \end{cases} \quad (36)$$

For external coordinate i , the transport equation is

$$\frac{dY_i}{dt} = S_i(t) + \sum_{e:d(e)=i} \lambda_e(t)Y_{s(e)} - Y_i \sum_{e:s(e)=i} \lambda_e(t). \quad (37)$$

The source, loss, and incoming-edge coefficient tables are evaluated at the 513 stored native times and linearly interpolated within each interval.

External nuclear coordinates are ordered by increasing (A, Z) . Every retained capture increases A , while every retained beta-minus edge preserves A and increases Z . The internal graph is therefore triangular and acyclic in this ordering. Outgoing reactions from a retained coordinate to a product above the chosen $A_{\max}$ are kept as outer-sink losses, but the unrepresented product is not added as a state.

The complete central $A = 24\text{--}119$ endpoint register contains 1,819 external graph coordinates. Of these, 1,817 have strictly positive 256-step endpoint populations and two— ^{45}Mg and ^{51}Si —are exactly zero. The validation claim is at the mass-sector level: every sum Y_A for $A = 24\text{--}119$ is strictly positive at every one of the six anchors.

Reverse photodisintegration, reaction topologies outside Equations (31) and (34), depletion of the prescribed light bath, depletion of native PRIMAT source species by the external calculation, and thermodynamic feedback from the external states are outside this operator. The resulting calculation is therefore a selected forward transport model rather than a complete heavy-element reaction network.

6 Positive Stiff Transport and Numerical Qualification

6.1 Error metric and source normalization

All reported solver comparisons use the symmetric relative metric implemented in the solver-comparison code,

$$\delta_{\text{rel}}(a, b) = \frac{|a - b|}{\max(|a|, |b|, 10^{-300})}. \quad (38)$$

The total injected external-source coordinate is

$$Q = \int_{t_0}^{t_f} \sum_i S_i(t) dt. \quad (39)$$

The production calculation is performed once with the normalized source S/Q for numerical conditioning and separately with the absolute source S . Multiplying the normalized endpoint by Q must reconstruct the absolute endpoint. This is a linearity/reconstruction control, not a normalization of the full native+external baryon inventory.

6.2 Positivity-preserving exponential-inflow update

For each stored native interval, the primary solver divides the interval into a fixed number of substeps. Coefficients are sampled by linear interpolation at the substep midpoint. Because incoming source coordinates precede their destinations in the triangular ordering, the upstream new value is already available when coordinate i is updated. Define

$$q_i^{m+1/2} = S_i^{m+1/2} + \sum_{e:d(e)=i} \lambda_e^{m+1/2} \frac{Y_{s(e)}^m + Y_{s(e)}^{m+1}}{2}, \quad (40)$$

and

$$L_i^{m+1/2} = \sum_{e:s(e)=i} \lambda_e^{m+1/2}. \quad (41)$$

The primary update is

$$Y_i^{m+1} = \begin{cases} \exp(-L_i \Delta t) Y_i^m + \frac{1 - \exp(-L_i \Delta t)}{L_i} q_i, & L_i > 0, \\ Y_i^m + \Delta t q_i, & L_i = 0, \end{cases} \quad (42)$$

where q_i and L_i in Equation (42) denote the midpoint values in Equation (40). The implementation evaluates the gain with `expm1` for numerical stability. With nonnegative old states, sources, and coefficients, Equation (42) preserves nonnegativity.

6.3 Independent triangular backward-Euler update

The verification solver samples the linearly interpolated coefficients at the right endpoint of each substep and applies

$$Y_i^{m+1} = \frac{Y_i^m + \Delta t S_i^{m+1} + \Delta t \sum_{e:d(e)=i} \lambda_e^{m+1} Y_{s(e)}^{m+1}}{1 + \Delta t \sum_{e:s(e)=i} \lambda_e^{m+1}}. \quad (43)$$

The triangular ordering again makes every required upstream $Y_{s(e)}^{m+1}$ available within the same substep and preserves nonnegative states for nonnegative coefficients.

6.4 Asymptotic continuum qualification

The production resolution is qualified independently of the $A = 1$ –119 endpoint table. The qualification uses 64, 128, and 256 substeps for both schemes at the eight preregistered checkpoint masses

$$\mathcal{A}_{\text{chk}} = \{24, 28, 32, 40, 56, 80, 100, 119\}. \quad (44)$$

For each anchor and method, let $D_{64,128}$ and $D_{128,256}$ be the maximum Equation (38) value over $\mathcal{A}_{\text{chk}}$. The reported aggregate effective order is

$$p_{\text{agg}} = \log_2 \left(\frac{D_{64,128}}{D_{128,256}} \right). \quad (45)$$

Separately, for each checkpoint mass A , the generalized three-level Richardson order and continuum estimate are

$$p_A = \log_2 \left(\frac{|Y_A^{64} - Y_A^{128}|}{|Y_A^{128} - Y_A^{256}|} \right), \quad (46)$$

$$Y_{A,\infty} = Y_A^{256} + \frac{Y_A^{256} - Y_A^{128}}{2^{p_A} - 1}. \quad (47)$$

The continuum cross-method metric is the maximum symmetric relative difference between the independently extrapolated primary and backward-Euler $Y_{A,\infty}$ values over $\mathcal{A}_{\text{chk}}$.

Table 3 gives the anchor-level checkpoint maxima. The primary per-checkpoint Richardson orders over all anchors range from 1.98656 to 2.00086; the backward-Euler per-checkpoint orders range from 0.99831 to 1.07400. Thus the individual checkpoint sequences show the same near-second-order/near-first-order structure indicated by the aggregate metrics.

η_{10}	Primary 128–256	Aggregate order	BE 128–256	Aggregate order	Continuum cross	256–Richardson
6.044	0.1875%	1.989457	3.1764%	0.975679	0.2068%	0.0632%
6.080	0.1885%	1.989396	3.1705%	0.975688	0.2068%	0.0636%
6.094	0.1889%	1.989372	3.1683%	0.975692	0.2068%	0.0637%
6.106	0.1892%	1.989352	3.1663%	0.975695	0.2068%	0.0638%
6.128	0.1899%	1.989315	3.1627%	0.975850	0.2067%	0.0640%
6.134	0.1901%	1.989305	3.1618%	0.976341	0.2067%	0.0641%

Table 3: Asymptotic qualification at the eight preregistered checkpoint masses. The “aggregate order” is Equation (45); Richardson limits are constructed separately mass by mass using Equation (47).

Across all six anchors, the worst primary 128–256 checkpoint difference is 0.190056%, the global aggregate primary order is 1.989305, the worst backward-Euler 128–256 checkpoint difference is 3.176411%, and the global aggregate backward-Euler order is 0.975679. The two independently extrapolated checkpoint continua disagree by at most 0.206809%. The 256-step primary checkpoint values are at most 0.064104% from their Richardson limits. These preregistered results qualify 256 substeps for the production calculation *at the declared checkpoint-based numerical-qualification scope*.

No heavy-sector endpoint output from another transport calculation and no asymptotic-qualification endpoint array is imported into the six-anchor production calculation.

7 Six-Anchor Validation

The six-anchor production run reconstructs the selected rates and graph at every anchor and then applies the frozen 256-substep primary scheme. Each anchor is required to have exactly fourteen boundary-source edges and 2,927 external states in the $A \leq 160$ graph; all six satisfy those identities. The full-state/all-sector hard gates require finite nonnegative states, no missing mass sector from $A = 24$ through 119, strictly positive mass-sector support over the same interval, exact zero-source response, full-vector source half/double linearity at 10^{-10} , and invariance of all $A = 24$ –119 mass-sector sums under the three outer graph limits $A_{\text{max}} = 128, 140, 160$. The separate absolute/normalized reconstruction check is evaluated at the eight checkpoint masses in Equation (44) and must also be below 10^{-10} .

Table 4 reports the anchor-dependent source integral and final controls. The last column is deliberately report-only: it compares the two finite-grid 256-step solutions at the eight checkpoint masses. Its $\sim 3.17\%$ scale is consistent with the visibly slower first-order backward-Euler sequence and is not the cross-method continuum gate; the independently extrapolated continua are compared in Table 3.

η_{10}	Boundary-source integral Q	Source linearity	Checkpoint reconstruction	Primary256-BE256
6.044	1.600457×10^{-36}	1.619×10^{-19}	3.422×10^{-13}	3.1851%
6.080	1.660383×10^{-36}	6.458×10^{-20}	1.685×10^{-13}	3.1793%
6.094	1.684193×10^{-36}	8.676×10^{-20}	2.687×10^{-13}	3.1771%
6.106	1.704831×10^{-36}	2.072×10^{-17}	3.486×10^{-13}	3.1751%
6.128	1.743222×10^{-36}	1.338×10^{-19}	9.658×10^{-14}	3.1716%
6.134	1.753818×10^{-36}	1.004×10^{-19}	2.004×10^{-13}	3.1707%

Table 4: Six-anchor production diagnostics. Every anchor separately has 14 boundary-source edges and 2,927 external states in the $A \leq 160$ graph. The reconstruction column and Primary256-BE256 are both eight-checkpoint diagnostics; the latter is report-only and is not the continuum-agreement metric.

Every anchor passes the frozen production gates. The worst source-linearity residual is 2.0723×10^{-17} , and the worst eight-checkpoint absolute/normalized reconstruction residual is 3.4863×10^{-13} . An exact-zero source gives an exact-zero external vector. Every mass-sector sum Y_A from $A = 24$ through 119 is finite and strictly positive at every anchor.

The validation independently rebuilds the graph with $A_{\max} = 128, 140, 160$. All $A \leq 119$ mass-sector results are identical in stored output, giving zero measured relative difference. This is an implementation-consistency test, not an independent physical-closure test: because the selected operator is strictly forward and acyclic, states above $A = 119$ cannot feed back into the reported region.

8 Mass-Sector Results

Table 5 gives representative mass sectors at the central $\eta_{10} = 6.094$ anchor together with the full six-anchor range. For $A \leq 23$, these are native PRIMAT C-backend final mass-sector sums. For $A \geq 24$, they are absolute post-processed external transport populations reconstructed from the prescribed source. The two classes are plotted together only to display scale and continuity of the register; they are not one self-consistent coupled-network endpoint vector. The six-anchor envelope is part of the reported computational result and is not an observational uncertainty distribution.

A	Central Y_A	Six-anchor minimum	Six-anchor maximum	Full range / central
1	7.530073×10^{-1}	7.52945×10^{-1}	7.53086×10^{-1}	0.019%
2	1.856299×10^{-5}	1.83633×10^{-5}	1.88172×10^{-5}	2.445%
3	7.86162×10^{-6}	7.8313×10^{-6}	7.90006×10^{-6}	0.875%
4	6.173301×10^{-2}	6.17131×10^{-2}	6.17488×10^{-2}	0.058%
7	4.080752×10^{-10}	4.01174×10^{-10}	4.13619×10^{-10}	3.050%
16	7.158061×10^{-20}	6.98152×10^{-20}	7.30122×10^{-20}	4.466%
23	1.195179×10^{-37}	1.15044×10^{-37}	1.23197×10^{-37}	6.821%
24	1.191108×10^{-36}	1.13334×10^{-36}	1.23907×10^{-36}	8.876%
28	1.133223×10^{-38}	1.06391×10^{-38}	1.19149×10^{-38}	11.258%
40	8.555989×10^{-46}	7.56013×10^{-46}	9.43835×10^{-46}	21.952%
56	1.485023×10^{-50}	1.24891×10^{-50}	1.70348×10^{-50}	30.610%
80	1.694513×10^{-53}	1.39764×10^{-53}	1.974×10^{-53}	34.014%
100	6.501512×10^{-56}	5.34969×10^{-56}	7.58855×10^{-56}	34.436%
113	1.656442×10^{-56}	1.35994×10^{-56}	1.93681×10^{-56}	34.826%
119	2.883556×10^{-56}	2.36803×10^{-56}	3.37091×10^{-56}	34.779%

Table 5: Selected mass-sector results. Masses $A \leq 23$ are PRIMAT-native; masses $A \geq 24$ are external reduced-graph results.

At the central anchor, the weakest external mass sector is $A = 113$, with $Y_{113} = 1.65644 \times 10^{-56}$. The final sector is

$$Y_{119}(\eta_{10} = 6.094) = 2.8835564 \times 10^{-56}, \quad (48)$$

with six-anchor range

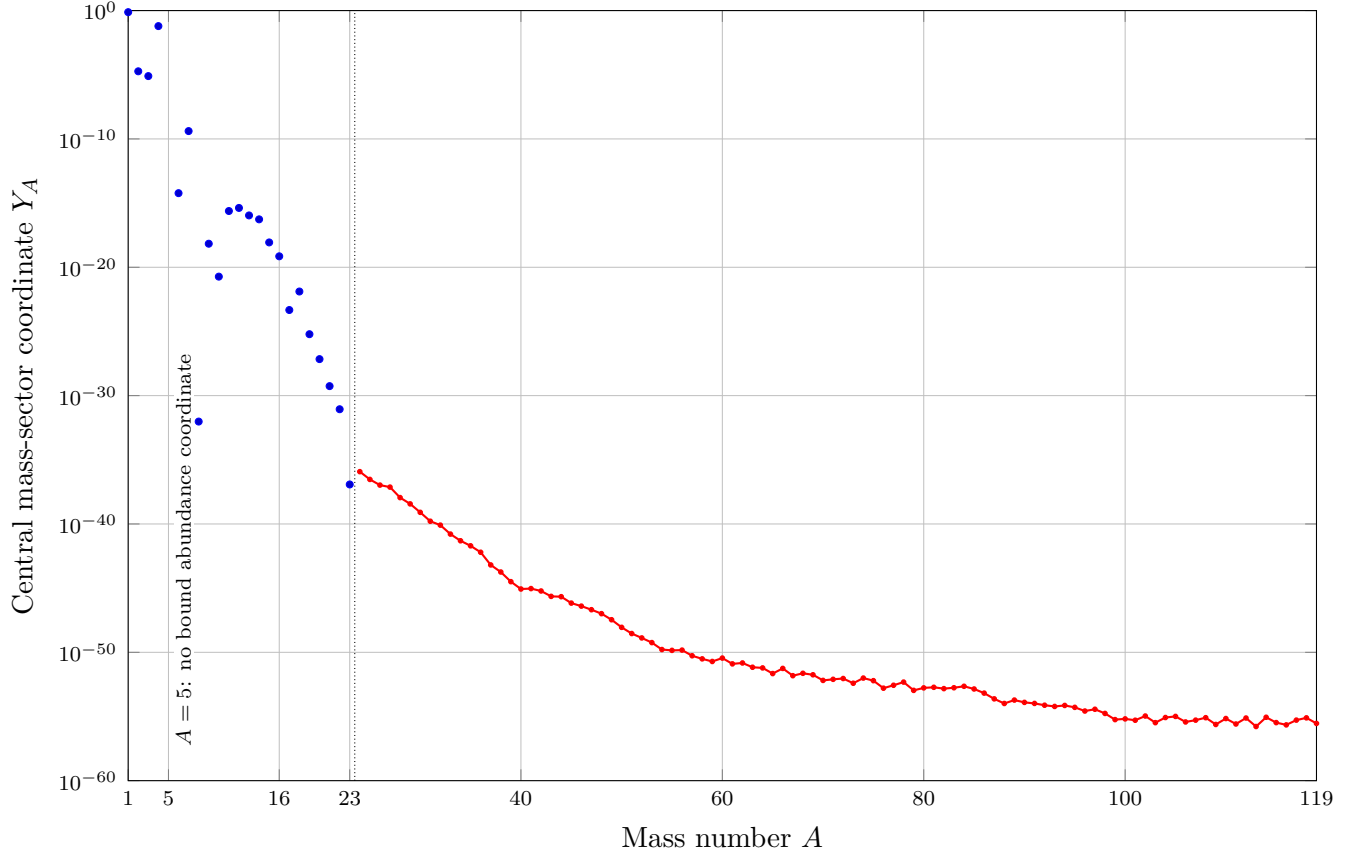
$$2.3680264 \times 10^{-56} \leq Y_{119} \leq 3.3709126 \times 10^{-56}. \quad (49)$$

The full range is approximately 34.779% of the central value. The largest corresponding external-sector range occurs at $A = 104$, where the six-anchor full range is approximately 35.397% of the central value. These variations reinforce the distinction between robust structural reachability and the amplitude of the deep tail.

9 Complete Mass-Sector Register from Mass 1 Through Mass 119

The complete register is generated directly from the six-anchor calculation. The native rows use the 59-species PRIMAT C-backend final inventory, grouped by mass through $A = 23$. The external rows list every graph coordinate in the central 256-step $A \leq 119$ endpoint register, comprising 1,819 external coordinates. At fixed mass number, different atomic numbers are listed as distinct isobaric nuclear coordinates; the free neutron is retained as its own native nuclear species.

The table has three class codes: **PN**, PRIMAT-native abundance coordinate; **PX**, prescribed-PRIMAT-bath external reduced-graph coordinate; and **G**, the non-abundance mass-five gap annotation. Central and envelope values are mass-sector sums, not individual-species values. For PX rows, the tabulated values are the absolute-source 256-step transport populations; the source-normalized coordinates are retained for numerical conditioning and controls but are not tabulated as the absolute-source register column. Graph membership also does not imply a positive endpoint for every listed coordinate: ^{45}Mg and ^{51}Si are exactly zero at the 256-step endpoint at all six anchors, while the corresponding mass-sector sums remain positive. Listed external nuclear coordinates are rate-library graph coordinates;



• PRIMAT-native, $A \leq 23$ — External reduced graph, $A = 24\text{--}119$

Figure 1: Central $\eta_{10} = 6.094$ mass-sector register. The vertical line marks the campaign's native/external boundary. Native PRIMAT endpoint sums and external post-processed transport populations are shown on the same ordinate for scale only; the combined plotted vector is not the endpoint of one coupled reaction network.

their appearance in the table is not an assertion of observational detection, stability, or measurable primordial production.

Printed digits reproduce the computed output and do not imply corresponding physical significant figures. In particular, the very small external values should be interpreted as numerical transport populations inside the declared operator, not as precision cosmological abundance predictions.

Table 6: Complete mass-sector register from $A = 1$ through $A = 119$. N is the number of modeled nuclear coordinates listed at each mass; the $A = 5$ resonance annotations are not counted as abundance coordinates. The six-anchor range is $[Y_{A,\min}, Y_{A,\max}]$.

A	N	Represented species / annotations	Y_A at $\eta_{10} = 6.094$	Six-anchor range	Class
1	2	n , ${}^1\text{H}$ (p)	7.530073×10^{-1}	$[7.52945 \times 10^{-1}, 7.53086 \times 10^{-1}]$	PN
2	1	${}^2\text{H}$ (D)	1.856299×10^{-5}	$[1.83633 \times 10^{-5}, 1.88172 \times 10^{-5}]$	PN
3	2	${}^3\text{H}$ (T), ${}^3\text{He}$	7.86162×10^{-6}	$[7.8313 \times 10^{-6}, 7.90006 \times 10^{-6}]$	PN
4	1	${}^4\text{He}$	6.173301×10^{-2}	$[6.17131 \times 10^{-2}, 6.17488 \times 10^{-2}]$	PN
5	0	No bound PRIMAT abundance coordinate; ${}^5\text{He}$ and ${}^5\text{Li}$ are unbound resonance annotations	not applicable	not applicable	G
6	2	${}^6\text{He}$, ${}^6\text{Li}$	5.945286×10^{-15}	$[5.88374 \times 10^{-15}, 6.0236 \times 10^{-15}]$	PN
7	2	${}^7\text{Li}$, ${}^7\text{Be}$	4.080752×10^{-10}	$[4.01174 \times 10^{-10}, 4.13619 \times 10^{-10}]$	PN
8	2	${}^8\text{Li}$, ${}^8\text{B}$	9.517439×10^{-33}	$[9.48615 \times 10^{-33}, 9.54267 \times 10^{-33}]$	PN
9	3	${}^9\text{Li}$, ${}^9\text{Be}$, ${}^9\text{C}$	6.851129×10^{-19}	$[6.7268 \times 10^{-19}, 7.01057 \times 10^{-19}]$	PN
10	3	${}^{10}\text{Be}$, ${}^{10}\text{B}$, ${}^{10}\text{C}$	1.867028×10^{-21}	$[1.84342 \times 10^{-21}, 1.89733 \times 10^{-21}]$	PN
11	3	${}^{11}\text{Be}$, ${}^{11}\text{B}$, ${}^{11}\text{C}$	2.416125×10^{-16}	$[2.33622 \times 10^{-16}, 2.48104 \times 10^{-16}]$	PN
12	4	${}^{12}\text{Be}$, ${}^{12}\text{B}$, ${}^{12}\text{C}$, ${}^{12}\text{N}$	4.202056×10^{-16}	$[4.19095 \times 10^{-16}, 4.21087 \times 10^{-16}]$	PN
13	4	${}^{13}\text{B}$, ${}^{13}\text{C}$, ${}^{13}\text{N}$, ${}^{13}\text{O}$	1.096271×10^{-16}	$[1.08694 \times 10^{-16}, 1.1037 \times 10^{-16}]$	PN
14	4	${}^{14}\text{B}$, ${}^{14}\text{C}$, ${}^{14}\text{N}$, ${}^{14}\text{O}$	5.422692×10^{-17}	$[5.31621 \times 10^{-17}, 5.50856 \times 10^{-17}]$	PN
15	4	${}^{15}\text{B}$, ${}^{15}\text{C}$, ${}^{15}\text{N}$, ${}^{15}\text{O}$	8.62551×10^{-19}	$[8.51187 \times 10^{-19}, 8.71654 \times 10^{-19}]$	PN
16	3	${}^{16}\text{C}$, ${}^{16}\text{N}$, ${}^{16}\text{O}$	7.158061×10^{-20}	$[6.98152 \times 10^{-20}, 7.30122 \times 10^{-20}]$	PN
17	3	${}^{17}\text{N}$, ${}^{17}\text{O}$, ${}^{17}\text{F}$	4.673344×10^{-24}	$[4.63913 \times 10^{-24}, 4.71619 \times 10^{-24}]$	PN
18	3	${}^{18}\text{O}$, ${}^{18}\text{F}$, ${}^{18}\text{Ne}$	1.293319×10^{-22}	$[1.24047 \times 10^{-22}, 1.33664 \times 10^{-22}]$	PN
19	3	${}^{19}\text{O}$, ${}^{19}\text{F}$, ${}^{19}\text{Ne}$	6.103579×10^{-26}	$[5.84881 \times 10^{-26}, 6.31287 \times 10^{-26}]$	PN
20	4	${}^{20}\text{O}$, ${}^{20}\text{F}$, ${}^{20}\text{Ne}$, ${}^{20}\text{Na}$	6.987941×10^{-28}	$[6.73305 \times 10^{-28}, 7.1973 \times 10^{-28}]$	PN
21	2	${}^{21}\text{Ne}$, ${}^{21}\text{Na}$	5.494649×10^{-30}	$[5.28927 \times 10^{-30}, 5.66339 \times 10^{-30}]$	PN
22	2	${}^{22}\text{Ne}$, ${}^{22}\text{Na}$	8.688187×10^{-32}	$[8.37861 \times 10^{-32}, 8.94153 \times 10^{-32}]$	PN
23	2	${}^{23}\text{Ne}$, ${}^{23}\text{Na}$	1.195179×10^{-37}	$[1.15044 \times 10^{-37}, 1.23197 \times 10^{-37}]$	PN
24	4	${}^{24}\text{Ne}$, ${}^{24}\text{Na}$, ${}^{24}\text{Mg}$, ${}^{24}\text{Al}$	1.191108×10^{-36}	$[1.13334 \times 10^{-36}, 1.23907 \times 10^{-36}]$	PX
25	5	${}^{25}\text{Ne}$, ${}^{25}\text{Na}$, ${}^{25}\text{Mg}$, ${}^{25}\text{Al}$, ${}^{25}\text{Si}$	2.957784×10^{-37}	$[2.80165 \times 10^{-37}, 3.08806 \times 10^{-37}]$	PX
26	6	${}^{26}\text{Ne}$, ${}^{26}\text{Na}$, ${}^{26}\text{Mg}$, ${}^{26}\text{Al}$, ${}^{26}\text{Si}$, ${}^{26}\text{P}$	1.067803×10^{-37}	$[1.01743 \times 10^{-37}, 1.10958 \times 10^{-37}]$	PX
27	7	${}^{27}\text{Ne}$, ${}^{27}\text{Na}$, ${}^{27}\text{Mg}$, ${}^{27}\text{Al}$, ${}^{27}\text{Si}$, ${}^{27}\text{P}$, ${}^{27}\text{S}$	7.445115×10^{-38}	$[7.01361 \times 10^{-38}, 7.80657 \times 10^{-38}]$	PX

Continued on next page

A	N	Represented species / annotations	Y_A at $\eta_{10} = 6.094$	Six-anchor range	Class
28	8	^{28}Ne , ^{28}Na , ^{28}Mg , ^{28}Al , ^{28}Si , ^{28}P , ^{28}S , ^{28}Cl	1.133223×10^{-38}	$[1.06391 \times 10^{-38}, 1.19149 \times 10^{-38}]$	PX
29	9	^{29}Ne , ^{29}Na , ^{29}Mg , ^{29}Al , ^{29}Si , ^{29}P , ^{29}S , ^{29}Cl , ^{29}Ar	3.671185×10^{-39}	$[3.4356 \times 10^{-39}, 3.86974 \times 10^{-39}]$	PX
30	10	^{30}Ne , ^{30}Na , ^{30}Mg , ^{30}Al , ^{30}Si , ^{30}P , ^{30}S , ^{30}Cl , ^{30}Ar , ^{30}K	8.043542×10^{-40}	$[7.49309 \times 10^{-40}, 8.50928 \times 10^{-40}]$	PX
31	11	^{31}Ne , ^{31}Na , ^{31}Mg , ^{31}Al , ^{31}Si , ^{31}P , ^{31}S , ^{31}Cl , ^{31}Ar , ^{31}K , ^{31}Ca	1.629293×10^{-40}	$[1.50837 \times 10^{-40}, 1.73216 \times 10^{-40}]$	PX
32	11	^{32}Ne , ^{32}Na , ^{32}Mg , ^{32}Al , ^{32}Si , ^{32}P , ^{32}S , ^{32}Cl , ^{32}Ar , ^{32}K , ^{32}Ca	8.107198×10^{-41}	$[7.47468 \times 10^{-41}, 8.64703 \times 10^{-41}]$	PX
33	12	^{33}Ne , ^{33}Na , ^{33}Mg , ^{33}Al , ^{33}Si , ^{33}P , ^{33}S , ^{33}Cl , ^{33}Ar , ^{33}K , ^{33}Ca , ^{33}Sc	1.624427×10^{-41}	$[1.48463 \times 10^{-41}, 1.74469 \times 10^{-41}]$	PX
34	12	^{34}Ne , ^{34}Na , ^{34}Mg , ^{34}Al , ^{34}Si , ^{34}P , ^{34}S , ^{34}Cl , ^{34}Ar , ^{34}K , ^{34}Ca , ^{34}Sc	4.908633×10^{-42}	$[4.47564 \times 10^{-42}, 5.28184 \times 10^{-42}]$	PX
35	13	^{35}Ne , ^{35}Na , ^{35}Mg , ^{35}Al , ^{35}Si , ^{35}P , ^{35}S , ^{35}Cl , ^{35}Ar , ^{35}K , ^{35}Ca , ^{35}Sc , ^{35}Ti	1.984364×10^{-42}	$[1.79667 \times 10^{-42}, 2.14712 \times 10^{-42}]$	PX
36	13	^{36}Ne , ^{36}Na , ^{36}Mg , ^{36}Al , ^{36}Si , ^{36}P , ^{36}S , ^{36}Cl , ^{36}Ar , ^{36}K , ^{36}Ca , ^{36}Sc , ^{36}Ti	6.249424×10^{-43}	$[5.64259 \times 10^{-43}, 6.77698 \times 10^{-43}]$	PX
37	14	^{37}Ne , ^{37}Na , ^{37}Mg , ^{37}Al , ^{37}Si , ^{37}P , ^{37}S , ^{37}Cl , ^{37}Ar , ^{37}K , ^{37}Ca , ^{37}Sc , ^{37}Ti , ^{37}V	6.436694×10^{-44}	$[5.75742 \times 10^{-44}, 7.03211 \times 10^{-44}]$	PX
38	14	^{38}Ne , ^{38}Na , ^{38}Mg , ^{38}Al , ^{38}Si , ^{38}P , ^{38}S , ^{38}Cl , ^{38}Ar , ^{38}K , ^{38}Ca , ^{38}Sc , ^{38}Ti , ^{38}V	1.806221×10^{-44}	$[1.60666 \times 10^{-44}, 1.98199 \times 10^{-44}]$	PX
39	14	^{39}Na , ^{39}Mg , ^{39}Al , ^{39}Si , ^{39}P , ^{39}S , ^{39}Cl , ^{39}Ar , ^{39}K , ^{39}Ca , ^{39}Sc , ^{39}Ti , ^{39}V , ^{39}Cr	3.201426×10^{-45}	$[2.83924 \times 10^{-45}, 3.52128 \times 10^{-45}]$	PX
40	14	^{40}Na , ^{40}Mg , ^{40}Al , ^{40}Si , ^{40}P , ^{40}S , ^{40}Cl , ^{40}Ar , ^{40}K , ^{40}Ca , ^{40}Sc , ^{40}Ti , ^{40}V , ^{40}Cr	8.555989×10^{-46}	$[7.56013 \times 10^{-46}, 9.43835 \times 10^{-46}]$	PX
41	15	^{41}Na , ^{41}Mg , ^{41}Al , ^{41}Si , ^{41}P , ^{41}S , ^{41}Cl , ^{41}Ar , ^{41}K , ^{41}Ca , ^{41}Sc , ^{41}Ti , ^{41}V , ^{41}Cr , ^{41}Mn	9.22856×10^{-46}	$[8.13285 \times 10^{-46}, 1.02015 \times 10^{-45}]$	PX
42	15	^{42}Na , ^{42}Mg , ^{42}Al , ^{42}Si , ^{42}P , ^{42}S , ^{42}Cl , ^{42}Ar , ^{42}K , ^{42}Ca , ^{42}Sc , ^{42}Ti , ^{42}V , ^{42}Cr , ^{42}Mn	5.998859×10^{-46}	$[5.27283 \times 10^{-46}, 6.64501 \times 10^{-46}]$	PX
43	15	^{43}Mg , ^{43}Al , ^{43}Si , ^{43}P , ^{43}S , ^{43}Cl , ^{43}Ar , ^{43}K , ^{43}Ca , ^{43}Sc , ^{43}Ti , ^{43}V , ^{43}Cr , ^{43}Mn , ^{43}Fe	2.280131×10^{-46}	$[1.99641 \times 10^{-46}, 2.53349 \times 10^{-46}]$	PX
44	15	^{44}Mg , ^{44}Al , ^{44}Si , ^{44}P , ^{44}S , ^{44}Cl , ^{44}Ar , ^{44}K , ^{44}Ca , ^{44}Sc , ^{44}Ti , ^{44}V , ^{44}Cr , ^{44}Mn , ^{44}Fe	2.158913×10^{-46}	$[1.88252 \times 10^{-46}, 2.4066 \times 10^{-46}]$	PX
45	16	^{45}Mg , ^{45}Al , ^{45}Si , ^{45}P , ^{45}S , ^{45}Cl , ^{45}Ar , ^{45}K , ^{45}Ca , ^{45}Sc , ^{45}Ti , ^{45}V , ^{45}Cr , ^{45}Mn , ^{45}Fe , ^{45}Co	6.825797×10^{-47}	$[5.93749 \times 10^{-47}, 7.62358 \times 10^{-47}]$	PX
46	15	^{46}Al , ^{46}Si , ^{46}P , ^{46}S , ^{46}Cl , ^{46}Ar , ^{46}K , ^{46}Ca , ^{46}Sc , ^{46}Ti , ^{46}V , ^{46}Cr , ^{46}Mn , ^{46}Fe , ^{46}Co	3.97234×10^{-47}	$[3.43792 \times 10^{-47}, 4.45444 \times 10^{-47}]$	PX
47	16	^{47}Al , ^{47}Si , ^{47}P , ^{47}S , ^{47}Cl , ^{47}Ar , ^{47}K , ^{47}Ca , ^{47}Sc , ^{47}Ti , ^{47}V , ^{47}Cr , ^{47}Mn , ^{47}Fe , ^{47}Co , ^{47}Ni	2.091534×10^{-47}	$[1.80388 \times 10^{-47}, 2.35181 \times 10^{-47}]$	PX
48	16	^{48}Al , ^{48}Si , ^{48}P , ^{48}S , ^{48}Cl , ^{48}Ar , ^{48}K , ^{48}Ca , ^{48}Sc , ^{48}Ti , ^{48}V , ^{48}Cr , ^{48}Mn , ^{48}Fe , ^{48}Co , ^{48}Ni	1.027738×10^{-47}	$[8.8367 \times 10^{-48}, 1.15845 \times 10^{-47}]$	PX
49	16	^{49}Si , ^{49}P , ^{49}S , ^{49}Cl , ^{49}Ar , ^{49}K , ^{49}Ca , ^{49}Sc , ^{49}Ti , ^{49}V , ^{49}Cr , ^{49}Mn , ^{49}Fe , ^{49}Co , ^{49}Ni , ^{49}Cu	3.457744×10^{-48}	$[2.95264 \times 10^{-48}, 3.91882 \times 10^{-48}]$	PX
50	16	^{50}Si , ^{50}P , ^{50}S , ^{50}Cl , ^{50}Ar , ^{50}K , ^{50}Ca , ^{50}Sc , ^{50}Ti , ^{50}V , ^{50}Cr , ^{50}Mn , ^{50}Fe , ^{50}Co , ^{50}Ni , ^{50}Cu	8.821852×10^{-49}	$[7.50282 \times 10^{-49}, 1.00302 \times 10^{-48}]$	PX

Continued on next page

A	N	Represented species / annotations	Y_A at $\eta_{10} = 6.094$	Six-anchor range	Class
51	16	^{51}Si , ^{51}P , ^{51}S , ^{51}Cl , ^{51}Ar , ^{51}K , ^{51}Ca , ^{51}Sc , ^{51}Ti , ^{51}V , ^{51}Cr , ^{51}Mn , ^{51}Fe , ^{51}Co , ^{51}Ni , ^{51}Cu	2.903424×10^{-49}	$[2.46214 \times 10^{-49}, 3.30875 \times 10^{-49}]$	PX
52	16	^{52}P , ^{52}S , ^{52}Cl , ^{52}Ar , ^{52}K , ^{52}Ca , ^{52}Sc , ^{52}Ti , ^{52}V , ^{52}Cr , ^{52}Mn , ^{52}Fe , ^{52}Co , ^{52}Ni , ^{52}Cu , ^{52}Zn	1.320659×10^{-49}	$[1.11836 \times 10^{-49}, 1.5067 \times 10^{-49}]$	PX
53	16	^{53}P , ^{53}S , ^{53}Cl , ^{53}Ar , ^{53}K , ^{53}Ca , ^{53}Sc , ^{53}Ti , ^{53}V , ^{53}Cr , ^{53}Mn , ^{53}Fe , ^{53}Co , ^{53}Ni , ^{53}Cu , ^{53}Zn	5.720908×10^{-50}	$[4.83028 \times 10^{-50}, 6.54212 \times 10^{-50}]$	PX
54	17	^{54}P , ^{54}S , ^{54}Cl , ^{54}Ar , ^{54}K , ^{54}Ca , ^{54}Sc , ^{54}Ti , ^{54}V , ^{54}Cr , ^{54}Mn , ^{54}Fe , ^{54}Co , ^{54}Ni , ^{54}Cu , ^{54}Zn , ^{54}Ga	1.644081×10^{-50}	$[1.39037 \times 10^{-50}, 1.87771 \times 10^{-50}]$	PX
55	17	^{55}P , ^{55}S , ^{55}Cl , ^{55}Ar , ^{55}K , ^{55}Ca , ^{55}Sc , ^{55}Ti , ^{55}V , ^{55}Cr , ^{55}Mn , ^{55}Fe , ^{55}Co , ^{55}Ni , ^{55}Cu , ^{55}Zn , ^{55}Ga	1.416941×10^{-50}	$[1.19307 \times 10^{-50}, 1.62387 \times 10^{-50}]$	PX
56	17	^{56}S , ^{56}Cl , ^{56}Ar , ^{56}K , ^{56}Ca , ^{56}Sc , ^{56}Ti , ^{56}V , ^{56}Cr , ^{56}Mn , ^{56}Fe , ^{56}Co , ^{56}Ni , ^{56}Cu , ^{56}Zn , ^{56}Ga , ^{56}Ge	1.485023×10^{-50}	$[1.24891 \times 10^{-50}, 1.70348 \times 10^{-50}]$	PX
57	17	^{57}S , ^{57}Cl , ^{57}Ar , ^{57}K , ^{57}Ca , ^{57}Sc , ^{57}Ti , ^{57}V , ^{57}Cr , ^{57}Mn , ^{57}Fe , ^{57}Co , ^{57}Ni , ^{57}Cu , ^{57}Zn , ^{57}Ga , ^{57}Ge	5.397628×10^{-51}	$[4.53383 \times 10^{-51}, 6.19774 \times 10^{-51}]$	PX
58	18	^{58}S , ^{58}Cl , ^{58}Ar , ^{58}K , ^{58}Ca , ^{58}Sc , ^{58}Ti , ^{58}V , ^{58}Cr , ^{58}Mn , ^{58}Fe , ^{58}Co , ^{58}Ni , ^{58}Cu , ^{58}Zn , ^{58}Ga , ^{58}Ge , ^{58}As	3.096501×10^{-51}	$[2.59855 \times 10^{-51}, 3.55814 \times 10^{-51}]$	PX
59	17	^{59}Cl , ^{59}Ar , ^{59}K , ^{59}Ca , ^{59}Sc , ^{59}Ti , ^{59}V , ^{59}Cr , ^{59}Mn , ^{59}Fe , ^{59}Co , ^{59}Ni , ^{59}Cu , ^{59}Zn , ^{59}Ga , ^{59}Ge , ^{59}As	1.935471×10^{-51}	$[1.62161 \times 10^{-51}, 2.22685 \times 10^{-51}]$	PX
60	18	^{60}Cl , ^{60}Ar , ^{60}K , ^{60}Ca , ^{60}Sc , ^{60}Ti , ^{60}V , ^{60}Cr , ^{60}Mn , ^{60}Fe , ^{60}Co , ^{60}Ni , ^{60}Cu , ^{60}Zn , ^{60}Ga , ^{60}Ge , ^{60}As , ^{60}Se	3.55465×10^{-51}	$[2.97682 \times 10^{-51}, 4.09131 \times 10^{-51}]$	PX
61	18	^{61}Cl , ^{61}Ar , ^{61}K , ^{61}Ca , ^{61}Sc , ^{61}Ti , ^{61}V , ^{61}Cr , ^{61}Mn , ^{61}Fe , ^{61}Co , ^{61}Ni , ^{61}Cu , ^{61}Zn , ^{61}Ga , ^{61}Ge , ^{61}As , ^{61}Se	1.257636×10^{-51}	$[1.05224 \times 10^{-51}, 1.44855 \times 10^{-51}]$	PX
62	18	^{62}Ar , ^{62}K , ^{62}Ca , ^{62}Sc , ^{62}Ti , ^{62}V , ^{62}Cr , ^{62}Mn , ^{62}Fe , ^{62}Co , ^{62}Ni , ^{62}Cu , ^{62}Zn , ^{62}Ga , ^{62}Ge , ^{62}As , ^{62}Se , ^{62}Br	1.477202×10^{-51}	$[1.23283 \times 10^{-51}, 1.70484 \times 10^{-51}]$	PX
63	18	^{63}Ar , ^{63}K , ^{63}Ca , ^{63}Sc , ^{63}Ti , ^{63}V , ^{63}Cr , ^{63}Mn , ^{63}Fe , ^{63}Co , ^{63}Ni , ^{63}Cu , ^{63}Zn , ^{63}Ga , ^{63}Ge , ^{63}As , ^{63}Se , ^{63}Br	6.857056×10^{-52}	$[5.72772 \times 10^{-52}, 7.90833 \times 10^{-52}]$	PX
64	19	^{64}Ar , ^{64}K , ^{64}Ca , ^{64}Sc , ^{64}Ti , ^{64}V , ^{64}Cr , ^{64}Mn , ^{64}Fe , ^{64}Co , ^{64}Ni , ^{64}Cu , ^{64}Zn , ^{64}Ga , ^{64}Ge , ^{64}As , ^{64}Se , ^{64}Br , ^{64}Kr	6.092421×10^{-52}	$[5.08646 \times 10^{-52}, 7.02929 \times 10^{-52}]$	PX
65	18	^{65}K , ^{65}Ca , ^{65}Sc , ^{65}Ti , ^{65}V , ^{65}Cr , ^{65}Mn , ^{65}Fe , ^{65}Co , ^{65}Ni , ^{65}Cu , ^{65}Zn , ^{65}Ga , ^{65}Ge , ^{65}As , ^{65}Se , ^{65}Br , ^{65}Kr	2.207962×10^{-52}	$[1.84111 \times 10^{-52}, 2.54998 \times 10^{-52}]$	PX
66	19	^{66}K , ^{66}Ca , ^{66}Sc , ^{66}Ti , ^{66}V , ^{66}Cr , ^{66}Mn , ^{66}Fe , ^{66}Co , ^{66}Ni , ^{66}Cu , ^{66}Zn , ^{66}Ga , ^{66}Ge , ^{66}As , ^{66}Se , ^{66}Br , ^{66}Kr , ^{66}Rb	5.559314×10^{-52}	$[4.62872 \times 10^{-52}, 6.42805 \times 10^{-52}]$	PX
67	19	^{67}K , ^{67}Ca , ^{67}Sc , ^{67}Ti , ^{67}V , ^{67}Cr , ^{67}Mn , ^{67}Fe , ^{67}Co , ^{67}Ni , ^{67}Cu , ^{67}Zn , ^{67}Ga , ^{67}Ge , ^{67}As , ^{67}Se , ^{67}Br , ^{67}Kr , ^{67}Rb	1.511317×10^{-52}	$[1.25611 \times 10^{-52}, 1.74993 \times 10^{-52}]$	PX
68	19	^{68}Ca , ^{68}Sc , ^{68}Ti , ^{68}V , ^{68}Cr , ^{68}Mn , ^{68}Fe , ^{68}Co , ^{68}Ni , ^{68}Cu , ^{68}Zn , ^{68}Ga , ^{68}Ge , ^{68}As , ^{68}Se , ^{68}Br , ^{68}Kr , ^{68}Rb , ^{68}Sr	2.310675×10^{-52}	$[1.91491 \times 10^{-52}, 2.68165 \times 10^{-52}]$	PX

Continued on next page

A	N	Represented species / annotations	Y_A at $\eta_{10} = 6.094$	Six-anchor range	Class
69	19	$^{69}\text{Ca}, ^{69}\text{Sc}, ^{69}\text{Ti}, ^{69}\text{V}, ^{69}\text{Cr}, ^{69}\text{Mn}, ^{69}\text{Fe}, ^{69}\text{Co}, ^{69}\text{Ni}, ^{69}\text{Cu}, ^{69}\text{Zn}, ^{69}\text{Ga}, ^{69}\text{Ge}, ^{69}\text{As}, ^{69}\text{Se}, ^{69}\text{Br}, ^{69}\text{Kr}, ^{69}\text{Rb}, ^{69}\text{Sr}$	1.778044×10^{-52}	$[1.4755 \times 10^{-52}, 2.06131 \times 10^{-52}]$	PX
70	20	$^{70}\text{Ca}, ^{70}\text{Sc}, ^{70}\text{Ti}, ^{70}\text{V}, ^{70}\text{Cr}, ^{70}\text{Mn}, ^{70}\text{Fe}, ^{70}\text{Co}, ^{70}\text{Ni}, ^{70}\text{Cu}, ^{70}\text{Zn}, ^{70}\text{Ga}, ^{70}\text{Ge}, ^{70}\text{As}, ^{70}\text{Se}, ^{70}\text{Br}, ^{70}\text{Kr}, ^{70}\text{Rb}, ^{70}\text{Sr}, ^{70}\text{Y}$	6.480022×10^{-53}	$[5.38849 \times 10^{-53}, 7.50029 \times 10^{-53}]$	PX
71	20	$^{71}\text{Ca}, ^{71}\text{Sc}, ^{71}\text{Ti}, ^{71}\text{V}, ^{71}\text{Cr}, ^{71}\text{Mn}, ^{71}\text{Fe}, ^{71}\text{Co}, ^{71}\text{Ni}, ^{71}\text{Cu}, ^{71}\text{Zn}, ^{71}\text{Ga}, ^{71}\text{Ge}, ^{71}\text{As}, ^{71}\text{Se}, ^{71}\text{Br}, ^{71}\text{Kr}, ^{71}\text{Rb}, ^{71}\text{Sr}, ^{71}\text{Y}$	7.865566×10^{-53}	$[6.52011 \times 10^{-53}, 9.12652 \times 10^{-53}]$	PX
72	20	$^{72}\text{Sc}, ^{72}\text{Ti}, ^{72}\text{V}, ^{72}\text{Cr}, ^{72}\text{Mn}, ^{72}\text{Fe}, ^{72}\text{Co}, ^{72}\text{Ni}, ^{72}\text{Cu}, ^{72}\text{Zn}, ^{72}\text{Ga}, ^{72}\text{Ge}, ^{72}\text{As}, ^{72}\text{Se}, ^{72}\text{Br}, ^{72}\text{Kr}, ^{72}\text{Rb}, ^{72}\text{Sr}, ^{72}\text{Y}, ^{72}\text{Zr}$	9.088239×10^{-53}	$[7.55407 \times 10^{-53}, 1.05228 \times 10^{-52}]$	PX
73	20	$^{73}\text{Sc}, ^{73}\text{Ti}, ^{73}\text{V}, ^{73}\text{Cr}, ^{73}\text{Mn}, ^{73}\text{Fe}, ^{73}\text{Co}, ^{73}\text{Ni}, ^{73}\text{Cu}, ^{73}\text{Zn}, ^{73}\text{Ga}, ^{73}\text{Ge}, ^{73}\text{As}, ^{73}\text{Se}, ^{73}\text{Br}, ^{73}\text{Kr}, ^{73}\text{Rb}, ^{73}\text{Sr}, ^{73}\text{Y}, ^{73}\text{Zr}$	3.894074×10^{-53}	$[3.22716 \times 10^{-53}, 4.51928 \times 10^{-53}]$	PX
74	21	$^{74}\text{Sc}, ^{74}\text{Ti}, ^{74}\text{V}, ^{74}\text{Cr}, ^{74}\text{Mn}, ^{74}\text{Fe}, ^{74}\text{Co}, ^{74}\text{Ni}, ^{74}\text{Cu}, ^{74}\text{Zn}, ^{74}\text{Ga}, ^{74}\text{Ge}, ^{74}\text{As}, ^{74}\text{Se}, ^{74}\text{Br}, ^{74}\text{Kr}, ^{74}\text{Rb}, ^{74}\text{Sr}, ^{74}\text{Y}, ^{74}\text{Zr}, ^{74}\text{Nb}$	9.858161×10^{-53}	$[8.16345 \times 10^{-53}, 1.14479 \times 10^{-52}]$	PX
75	20	$^{75}\text{Ti}, ^{75}\text{V}, ^{75}\text{Cr}, ^{75}\text{Mn}, ^{75}\text{Fe}, ^{75}\text{Co}, ^{75}\text{Ni}, ^{75}\text{Cu}, ^{75}\text{Zn}, ^{75}\text{Ga}, ^{75}\text{Ge}, ^{75}\text{As}, ^{75}\text{Se}, ^{75}\text{Br}, ^{75}\text{Kr}, ^{75}\text{Rb}, ^{75}\text{Sr}, ^{75}\text{Y}, ^{75}\text{Zr}, ^{75}\text{Nb}$	6.188862×10^{-53}	$[5.13537 \times 10^{-53}, 7.17545 \times 10^{-53}]$	PX
76	21	$^{76}\text{Ti}, ^{76}\text{V}, ^{76}\text{Cr}, ^{76}\text{Mn}, ^{76}\text{Fe}, ^{76}\text{Co}, ^{76}\text{Ni}, ^{76}\text{Cu}, ^{76}\text{Zn}, ^{76}\text{Ga}, ^{76}\text{Ge}, ^{76}\text{As}, ^{76}\text{Se}, ^{76}\text{Br}, ^{76}\text{Kr}, ^{76}\text{Rb}, ^{76}\text{Sr}, ^{76}\text{Y}, ^{76}\text{Zr}, ^{76}\text{Nb}, ^{76}\text{Mo}$	1.603237×10^{-53}	$[1.3315 \times 10^{-53}, 1.85756 \times 10^{-53}]$	PX
77	21	$^{77}\text{Ti}, ^{77}\text{V}, ^{77}\text{Cr}, ^{77}\text{Mn}, ^{77}\text{Fe}, ^{77}\text{Co}, ^{77}\text{Ni}, ^{77}\text{Cu}, ^{77}\text{Zn}, ^{77}\text{Ga}, ^{77}\text{Ge}, ^{77}\text{As}, ^{77}\text{Se}, ^{77}\text{Br}, ^{77}\text{Kr}, ^{77}\text{Rb}, ^{77}\text{Sr}, ^{77}\text{Y}, ^{77}\text{Zr}, ^{77}\text{Nb}, ^{77}\text{Mo}$	2.708816×10^{-53}	$[2.2374 \times 10^{-53}, 3.15206 \times 10^{-53}]$	PX
78	21	$^{78}\text{V}, ^{78}\text{Cr}, ^{78}\text{Mn}, ^{78}\text{Fe}, ^{78}\text{Co}, ^{78}\text{Ni}, ^{78}\text{Cu}, ^{78}\text{Zn}, ^{78}\text{Ga}, ^{78}\text{Ge}, ^{78}\text{As}, ^{78}\text{Se}, ^{78}\text{Br}, ^{78}\text{Kr}, ^{78}\text{Rb}, ^{78}\text{Sr}, ^{78}\text{Y}, ^{78}\text{Zr}, ^{78}\text{Nb}, ^{78}\text{Mo}, ^{78}\text{Tc}$	4.779516×10^{-53}	$[3.95737 \times 10^{-53}, 5.55094 \times 10^{-53}]$	PX
79	21	$^{79}\text{V}, ^{79}\text{Cr}, ^{79}\text{Mn}, ^{79}\text{Fe}, ^{79}\text{Co}, ^{79}\text{Ni}, ^{79}\text{Cu}, ^{79}\text{Zn}, ^{79}\text{Ga}, ^{79}\text{Ge}, ^{79}\text{As}, ^{79}\text{Se}, ^{79}\text{Br}, ^{79}\text{Kr}, ^{79}\text{Rb}, ^{79}\text{Sr}, ^{79}\text{Y}, ^{79}\text{Zr}, ^{79}\text{Nb}, ^{79}\text{Mo}, ^{79}\text{Tc}$	1.084229×10^{-53}	$[8.95627 \times 10^{-54}, 1.26156 \times 10^{-53}]$	PX
80	22	$^{80}\text{V}, ^{80}\text{Cr}, ^{80}\text{Mn}, ^{80}\text{Fe}, ^{80}\text{Co}, ^{80}\text{Ni}, ^{80}\text{Cu}, ^{80}\text{Zn}, ^{80}\text{Ga}, ^{80}\text{Ge}, ^{80}\text{As}, ^{80}\text{Se}, ^{80}\text{Br}, ^{80}\text{Kr}, ^{80}\text{Rb}, ^{80}\text{Sr}, ^{80}\text{Y}, ^{80}\text{Zr}, ^{80}\text{Nb}, ^{80}\text{Mo}, ^{80}\text{Tc}, ^{80}\text{Ru}$	1.694513×10^{-53}	$[1.39764 \times 10^{-53}, 1.974 \times 10^{-53}]$	PX
81	21	$^{81}\text{Cr}, ^{81}\text{Mn}, ^{81}\text{Fe}, ^{81}\text{Co}, ^{81}\text{Ni}, ^{81}\text{Cu}, ^{81}\text{Zn}, ^{81}\text{Ga}, ^{81}\text{Ge}, ^{81}\text{As}, ^{81}\text{Se}, ^{81}\text{Br}, ^{81}\text{Kr}, ^{81}\text{Rb}, ^{81}\text{Sr}, ^{81}\text{Y}, ^{81}\text{Zr}, ^{81}\text{Nb}, ^{81}\text{Mo}, ^{81}\text{Tc}, ^{81}\text{Ru}$	1.878057×10^{-53}	$[1.5498 \times 10^{-53}, 2.18696 \times 10^{-53}]$	PX
82	22	$^{82}\text{Cr}, ^{82}\text{Mn}, ^{82}\text{Fe}, ^{82}\text{Co}, ^{82}\text{Ni}, ^{82}\text{Cu}, ^{82}\text{Zn}, ^{82}\text{Ga}, ^{82}\text{Ge}, ^{82}\text{As}, ^{82}\text{Se}, ^{82}\text{Br}, ^{82}\text{Kr}, ^{82}\text{Rb}, ^{82}\text{Sr}, ^{82}\text{Y}, ^{82}\text{Zr}, ^{82}\text{Nb}, ^{82}\text{Mo}, ^{82}\text{Tc}, ^{82}\text{Ru}, ^{82}\text{Rh}$	1.48309×10^{-53}	$[1.22814 \times 10^{-53}, 1.72231 \times 10^{-53}]$	PX
83	22	$^{83}\text{Cr}, ^{83}\text{Mn}, ^{83}\text{Fe}, ^{83}\text{Co}, ^{83}\text{Ni}, ^{83}\text{Cu}, ^{83}\text{Zn}, ^{83}\text{Ga}, ^{83}\text{Ge}, ^{83}\text{As}, ^{83}\text{Se}, ^{83}\text{Br}, ^{83}\text{Kr}, ^{83}\text{Rb}, ^{83}\text{Sr}, ^{83}\text{Y}, ^{83}\text{Zr}, ^{83}\text{Nb}, ^{83}\text{Mo}, ^{83}\text{Tc}, ^{83}\text{Ru}, ^{83}\text{Rh}$	1.736685×10^{-53}	$[1.43086 \times 10^{-53}, 2.02491 \times 10^{-53}]$	PX
84	23	$^{84}\text{Cr}, ^{84}\text{Mn}, ^{84}\text{Fe}, ^{84}\text{Co}, ^{84}\text{Ni}, ^{84}\text{Cu}, ^{84}\text{Zn}, ^{84}\text{Ga}, ^{84}\text{Ge}, ^{84}\text{As}, ^{84}\text{Se}, ^{84}\text{Br}, ^{84}\text{Kr}, ^{84}\text{Rb}, ^{84}\text{Sr}, ^{84}\text{Y}, ^{84}\text{Zr}, ^{84}\text{Nb}, ^{84}\text{Mo}, ^{84}\text{Tc}, ^{84}\text{Ru}, ^{84}\text{Rh}, ^{84}\text{Pd}$	2.254238×10^{-53}	$[1.86028 \times 10^{-53}, 2.62501 \times 10^{-53}]$	PX
85	22	$^{85}\text{Mn}, ^{85}\text{Fe}, ^{85}\text{Co}, ^{85}\text{Ni}, ^{85}\text{Cu}, ^{85}\text{Zn}, ^{85}\text{Ga}, ^{85}\text{Ge}, ^{85}\text{As}, ^{85}\text{Se}, ^{85}\text{Br}, ^{85}\text{Kr}, ^{85}\text{Rb}, ^{85}\text{Sr}, ^{85}\text{Y}, ^{85}\text{Zr}, ^{85}\text{Nb}, ^{85}\text{Mo}, ^{85}\text{Tc}, ^{85}\text{Ru}, ^{85}\text{Rh}, ^{85}\text{Pd}$	1.388467×10^{-53}	$[1.14536 \times 10^{-53}, 1.61737 \times 10^{-53}]$	PX

Continued on next page

A	N	Represented species / annotations	Y_A at $\eta_{10} = 6.094$	Six-anchor range	Class
86	23	$^{86}\text{Mn}, ^{86}\text{Fe}, ^{86}\text{Co}, ^{86}\text{Ni}, ^{86}\text{Cu}, ^{86}\text{Zn}, ^{86}\text{Ga}, ^{86}\text{Ge}, ^{86}\text{As}, ^{86}\text{Se}, ^{86}\text{Br}, ^{86}\text{Kr}, ^{86}\text{Rb}, ^{86}\text{Sr}, ^{86}\text{Y}, ^{86}\text{Zr}, ^{86}\text{Nb}, ^{86}\text{Mo}, ^{86}\text{Tc}, ^{86}\text{Ru}, ^{86}\text{Rh}, ^{86}\text{Pd}, ^{86}\text{Ag}$	6.617346×10^{-54}	$[5.45901 \times 10^{-54}, 7.70797 \times 10^{-54}]$	PX
87	23	$^{87}\text{Mn}, ^{87}\text{Fe}, ^{87}\text{Co}, ^{87}\text{Ni}, ^{87}\text{Cu}, ^{87}\text{Zn}, ^{87}\text{Ga}, ^{87}\text{Ge}, ^{87}\text{As}, ^{87}\text{Se}, ^{87}\text{Br}, ^{87}\text{Kr}, ^{87}\text{Rb}, ^{87}\text{Sr}, ^{87}\text{Y}, ^{87}\text{Zr}, ^{87}\text{Nb}, ^{87}\text{Mo}, ^{87}\text{Tc}, ^{87}\text{Ru}, ^{87}\text{Rh}, ^{87}\text{Pd}, ^{87}\text{Ag}$	2.36441×10^{-54}	$[1.94784 \times 10^{-54}, 2.75709 \times 10^{-54}]$	PX
88	23	$^{88}\text{Fe}, ^{88}\text{Co}, ^{88}\text{Ni}, ^{88}\text{Cu}, ^{88}\text{Zn}, ^{88}\text{Ga}, ^{88}\text{Ge}, ^{88}\text{As}, ^{88}\text{Se}, ^{88}\text{Br}, ^{88}\text{Kr}, ^{88}\text{Rb}, ^{88}\text{Sr}, ^{88}\text{Y}, ^{88}\text{Zr}, ^{88}\text{Nb}, ^{88}\text{Mo}, ^{88}\text{Tc}, ^{88}\text{Ru}, ^{88}\text{Rh}, ^{88}\text{Pd}, ^{88}\text{Ag}, ^{88}\text{Cd}$	1.032663×10^{-54}	$[8.52188 \times 10^{-55}, 1.20254 \times 10^{-54}]$	PX
89	23	$^{89}\text{Fe}, ^{89}\text{Co}, ^{89}\text{Ni}, ^{89}\text{Cu}, ^{89}\text{Zn}, ^{89}\text{Ga}, ^{89}\text{Ge}, ^{89}\text{As}, ^{89}\text{Se}, ^{89}\text{Br}, ^{89}\text{Kr}, ^{89}\text{Rb}, ^{89}\text{Sr}, ^{89}\text{Y}, ^{89}\text{Zr}, ^{89}\text{Nb}, ^{89}\text{Mo}, ^{89}\text{Tc}, ^{89}\text{Ru}, ^{89}\text{Rh}, ^{89}\text{Pd}, ^{89}\text{Ag}, ^{89}\text{Cd}$	1.900742×10^{-54}	$[1.56491 \times 10^{-54}, 2.21748 \times 10^{-54}]$	PX
90	24	$^{90}\text{Fe}, ^{90}\text{Co}, ^{90}\text{Ni}, ^{90}\text{Cu}, ^{90}\text{Zn}, ^{90}\text{Ga}, ^{90}\text{Ge}, ^{90}\text{As}, ^{90}\text{Se}, ^{90}\text{Br}, ^{90}\text{Kr}, ^{90}\text{Rb}, ^{90}\text{Sr}, ^{90}\text{Y}, ^{90}\text{Zr}, ^{90}\text{Nb}, ^{90}\text{Mo}, ^{90}\text{Tc}, ^{90}\text{Ru}, ^{90}\text{Rh}, ^{90}\text{Pd}, ^{90}\text{Ag}, ^{90}\text{Cd}, ^{90}\text{In}$	1.257316×10^{-54}	$[1.0338 \times 10^{-54}, 1.46837 \times 10^{-54}]$	PX
91	23	$^{91}\text{Co}, ^{91}\text{Ni}, ^{91}\text{Cu}, ^{91}\text{Zn}, ^{91}\text{Ga}, ^{91}\text{Ge}, ^{91}\text{As}, ^{91}\text{Se}, ^{91}\text{Br}, ^{91}\text{Kr}, ^{91}\text{Rb}, ^{91}\text{Sr}, ^{91}\text{Y}, ^{91}\text{Zr}, ^{91}\text{Nb}, ^{91}\text{Mo}, ^{91}\text{Tc}, ^{91}\text{Ru}, ^{91}\text{Rh}, ^{91}\text{Pd}, ^{91}\text{Ag}, ^{91}\text{Cd}, ^{91}\text{In}$	1.04162×10^{-54}	$[8.57619 \times 10^{-55}, 1.21516 \times 10^{-54}]$	PX
92	24	$^{92}\text{Co}, ^{92}\text{Ni}, ^{92}\text{Cu}, ^{92}\text{Zn}, ^{92}\text{Ga}, ^{92}\text{Ge}, ^{92}\text{As}, ^{92}\text{Se}, ^{92}\text{Br}, ^{92}\text{Kr}, ^{92}\text{Rb}, ^{92}\text{Sr}, ^{92}\text{Y}, ^{92}\text{Zr}, ^{92}\text{Nb}, ^{92}\text{Mo}, ^{92}\text{Tc}, ^{92}\text{Ru}, ^{92}\text{Rh}, ^{92}\text{Pd}, ^{92}\text{Ag}, ^{92}\text{Cd}, ^{92}\text{In}, ^{92}\text{Sn}$	7.545127×10^{-55}	$[6.21907 \times 10^{-55}, 8.79468 \times 10^{-55}]$	PX
93	24	$^{93}\text{Co}, ^{93}\text{Ni}, ^{93}\text{Cu}, ^{93}\text{Zn}, ^{93}\text{Ga}, ^{93}\text{Ge}, ^{93}\text{As}, ^{93}\text{Se}, ^{93}\text{Br}, ^{93}\text{Kr}, ^{93}\text{Rb}, ^{93}\text{Sr}, ^{93}\text{Y}, ^{93}\text{Zr}, ^{93}\text{Nb}, ^{93}\text{Mo}, ^{93}\text{Tc}, ^{93}\text{Ru}, ^{93}\text{Rh}, ^{93}\text{Pd}, ^{93}\text{Ag}, ^{93}\text{Cd}, ^{93}\text{In}, ^{93}\text{Sn}$	6.063009×10^{-55}	$[4.98595 \times 10^{-55}, 7.07993 \times 10^{-55}]$	PX
94	23	$^{94}\text{Ni}, ^{94}\text{Cu}, ^{94}\text{Zn}, ^{94}\text{Ga}, ^{94}\text{Ge}, ^{94}\text{As}, ^{94}\text{Se}, ^{94}\text{Br}, ^{94}\text{Kr}, ^{94}\text{Rb}, ^{94}\text{Sr}, ^{94}\text{Y}, ^{94}\text{Zr}, ^{94}\text{Nb}, ^{94}\text{Mo}, ^{94}\text{Tc}, ^{94}\text{Ru}, ^{94}\text{Rh}, ^{94}\text{Pd}, ^{94}\text{Ag}, ^{94}\text{Cd}, ^{94}\text{In}, ^{94}\text{Sn}$	7.238408×10^{-55}	$[5.95008 \times 10^{-55}, 8.45519 \times 10^{-55}]$	PX
95	23	$^{95}\text{Ni}, ^{95}\text{Cu}, ^{95}\text{Zn}, ^{95}\text{Ga}, ^{95}\text{Ge}, ^{95}\text{As}, ^{95}\text{Se}, ^{95}\text{Br}, ^{95}\text{Kr}, ^{95}\text{Rb}, ^{95}\text{Sr}, ^{95}\text{Y}, ^{95}\text{Zr}, ^{95}\text{Nb}, ^{95}\text{Mo}, ^{95}\text{Tc}, ^{95}\text{Ru}, ^{95}\text{Rh}, ^{95}\text{Pd}, ^{95}\text{Ag}, ^{95}\text{Cd}, ^{95}\text{In}, ^{95}\text{Sn}$	5.149812×10^{-55}	$[4.24108 \times 10^{-55}, 6.00677 \times 10^{-55}]$	PX
96	23	$^{96}\text{Ni}, ^{96}\text{Cu}, ^{96}\text{Zn}, ^{96}\text{Ga}, ^{96}\text{Ge}, ^{96}\text{As}, ^{96}\text{Se}, ^{96}\text{Br}, ^{96}\text{Kr}, ^{96}\text{Rb}, ^{96}\text{Sr}, ^{96}\text{Y}, ^{96}\text{Zr}, ^{96}\text{Nb}, ^{96}\text{Mo}, ^{96}\text{Tc}, ^{96}\text{Ru}, ^{96}\text{Rh}, ^{96}\text{Pd}, ^{96}\text{Ag}, ^{96}\text{Cd}, ^{96}\text{In}, ^{96}\text{Sn}$	2.664747×10^{-55}	$[2.19713 \times 10^{-55}, 3.10529 \times 10^{-55}]$	PX
97	22	$^{97}\text{Cu}, ^{97}\text{Zn}, ^{97}\text{Ga}, ^{97}\text{Ge}, ^{97}\text{As}, ^{97}\text{Se}, ^{97}\text{Br}, ^{97}\text{Kr}, ^{97}\text{Rb}, ^{97}\text{Sr}, ^{97}\text{Y}, ^{97}\text{Zr}, ^{97}\text{Nb}, ^{97}\text{Mo}, ^{97}\text{Tc}, ^{97}\text{Ru}, ^{97}\text{Rh}, ^{97}\text{Pd}, ^{97}\text{Ag}, ^{97}\text{Cd}, ^{97}\text{In}, ^{97}\text{Sn}$	3.657056×10^{-55}	$[3.008 \times 10^{-55}, 4.26978 \times 10^{-55}]$	PX
98	23	$^{98}\text{Cu}, ^{98}\text{Zn}, ^{98}\text{Ga}, ^{98}\text{Ge}, ^{98}\text{As}, ^{98}\text{Se}, ^{98}\text{Br}, ^{98}\text{Kr}, ^{98}\text{Rb}, ^{98}\text{Sr}, ^{98}\text{Y}, ^{98}\text{Zr}, ^{98}\text{Nb}, ^{98}\text{Mo}, ^{98}\text{Tc}, ^{98}\text{Ru}, ^{98}\text{Rh}, ^{98}\text{Pd}, ^{98}\text{Ag}, ^{98}\text{Cd}, ^{98}\text{In}, ^{98}\text{Sn}, ^{98}\text{Sb}$	1.729265×10^{-55}	$[1.41995 \times 10^{-55}, 2.02168 \times 10^{-55}]$	PX
99	23	$^{99}\text{Cu}, ^{99}\text{Zn}, ^{99}\text{Ga}, ^{99}\text{Ge}, ^{99}\text{As}, ^{99}\text{Se}, ^{99}\text{Br}, ^{99}\text{Kr}, ^{99}\text{Rb}, ^{99}\text{Sr}, ^{99}\text{Y}, ^{99}\text{Zr}, ^{99}\text{Nb}, ^{99}\text{Mo}, ^{99}\text{Tc}, ^{99}\text{Ru}, ^{99}\text{Rh}, ^{99}\text{Pd}, ^{99}\text{Ag}, ^{99}\text{Cd}, ^{99}\text{In}, ^{99}\text{Sn}, ^{99}\text{Sb}$	5.720434×10^{-56}	$[4.70357 \times 10^{-56}, 6.68064 \times 10^{-56}]$	PX
100	24	$^{100}\text{Cu}, ^{100}\text{Zn}, ^{100}\text{Ga}, ^{100}\text{Ge}, ^{100}\text{As}, ^{100}\text{Se}, ^{100}\text{Br}, ^{100}\text{Kr}, ^{100}\text{Rb}, ^{100}\text{Sr}, ^{100}\text{Y}, ^{100}\text{Zr}, ^{100}\text{Nb}, ^{100}\text{Mo}, ^{100}\text{Tc}, ^{100}\text{Ru}, ^{100}\text{Rh}, ^{100}\text{Pd}, ^{100}\text{Ag}, ^{100}\text{Cd}, ^{100}\text{In}, ^{100}\text{Sn}, ^{100}\text{Sb}, ^{100}\text{Te}$	6.501512×10^{-56}	$[5.34969 \times 10^{-56}, 7.58855 \times 10^{-56}]$	PX
101	23	$^{101}\text{Zn}, ^{101}\text{Ga}, ^{101}\text{Ge}, ^{101}\text{As}, ^{101}\text{Se}, ^{101}\text{Br}, ^{101}\text{Kr}, ^{101}\text{Rb}, ^{101}\text{Sr}, ^{101}\text{Y}, ^{101}\text{Zr}, ^{101}\text{Nb}, ^{101}\text{Mo}, ^{101}\text{Tc}, ^{101}\text{Ru}, ^{101}\text{Rh}, ^{101}\text{Pd}, ^{101}\text{Ag}, ^{101}\text{Cd}, ^{101}\text{In}, ^{101}\text{Sn}, ^{101}\text{Sb}, ^{101}\text{Te}$	5.10794×10^{-56}	$[4.19889 \times 10^{-56}, 5.96659 \times 10^{-56}]$	PX
102	24	$^{102}\text{Zn}, ^{102}\text{Ga}, ^{102}\text{Ge}, ^{102}\text{As}, ^{102}\text{Se}, ^{102}\text{Br}, ^{102}\text{Kr}, ^{102}\text{Rb}, ^{102}\text{Sr}, ^{102}\text{Y}, ^{102}\text{Zr}, ^{102}\text{Nb}, ^{102}\text{Mo}, ^{102}\text{Tc}, ^{102}\text{Ru}, ^{102}\text{Rh}, ^{102}\text{Pd}, ^{102}\text{Ag}, ^{102}\text{Cd}, ^{102}\text{In}, ^{102}\text{Sn}, ^{102}\text{Sb}, ^{102}\text{Te}, ^{102}\text{I}$	1.084926×10^{-55}	$[8.92767 \times 10^{-56}, 1.26627 \times 10^{-55}]$	PX

Continued on next page

A	N	Represented species / annotations	Y_A at $\eta_{10} = 6.094$	Six-anchor range	Class
103	24	$^{103}\text{Zn}, ^{103}\text{Ga}, ^{103}\text{Ge}, ^{103}\text{As}, ^{103}\text{Se}, ^{103}\text{Br}, ^{103}\text{Kr}, ^{103}\text{Rb}, ^{103}\text{Sr}, ^{103}\text{Y}, ^{103}\text{Zr}, ^{103}\text{Nb}, ^{103}\text{Mo}, ^{103}\text{Tc}, ^{103}\text{Ru}, ^{103}\text{Rh}, ^{103}\text{Pd}, ^{103}\text{Ag}, ^{103}\text{Cd}, ^{103}\text{In}, ^{103}\text{Sn}, ^{103}\text{Sb}, ^{103}\text{Te}, ^{103}\text{I}$	3.311411×10^{-56}	$[2.7191 \times 10^{-56}, 3.87142 \times 10^{-56}]$	PX
104	24	$^{104}\text{Ga}, ^{104}\text{Ge}, ^{104}\text{As}, ^{104}\text{Se}, ^{104}\text{Br}, ^{104}\text{Kr}, ^{104}\text{Rb}, ^{104}\text{Sr}, ^{104}\text{Y}, ^{104}\text{Zr}, ^{104}\text{Nb}, ^{104}\text{Mo}, ^{104}\text{Tc}, ^{104}\text{Ru}, ^{104}\text{Rh}, ^{104}\text{Pd}, ^{104}\text{Ag}, ^{104}\text{Cd}, ^{104}\text{In}, ^{104}\text{Sn}, ^{104}\text{Sb}, ^{104}\text{Te}, ^{104}\text{I}, ^{104}\text{Xe}$	8.314303×10^{-56}	$[6.80365 \times 10^{-56}, 9.74665 \times 10^{-56}]$	PX
105	24	$^{105}\text{Ga}, ^{105}\text{Ge}, ^{105}\text{As}, ^{105}\text{Se}, ^{105}\text{Br}, ^{105}\text{Kr}, ^{105}\text{Rb}, ^{105}\text{Sr}, ^{105}\text{Y}, ^{105}\text{Zr}, ^{105}\text{Nb}, ^{105}\text{Mo}, ^{105}\text{Tc}, ^{105}\text{Ru}, ^{105}\text{Rh}, ^{105}\text{Pd}, ^{105}\text{Ag}, ^{105}\text{Cd}, ^{105}\text{In}, ^{105}\text{Sn}, ^{105}\text{Sb}, ^{105}\text{Te}, ^{105}\text{I}, ^{105}\text{Xe}$	1.012846×10^{-55}	$[8.33279 \times 10^{-56}, 1.18232 \times 10^{-55}]$	PX
106	24	$^{106}\text{Ga}, ^{106}\text{Ge}, ^{106}\text{As}, ^{106}\text{Se}, ^{106}\text{Br}, ^{106}\text{Kr}, ^{106}\text{Rb}, ^{106}\text{Sr}, ^{106}\text{Y}, ^{106}\text{Zr}, ^{106}\text{Nb}, ^{106}\text{Mo}, ^{106}\text{Tc}, ^{106}\text{Ru}, ^{106}\text{Rh}, ^{106}\text{Pd}, ^{106}\text{Ag}, ^{106}\text{Cd}, ^{106}\text{In}, ^{106}\text{Sn}, ^{106}\text{Sb}, ^{106}\text{Te}, ^{106}\text{I}, ^{106}\text{Xe}$	3.753494×10^{-56}	$[3.08414 \times 10^{-56}, 4.38605 \times 10^{-56}]$	PX
107	24	$^{107}\text{Ge}, ^{107}\text{As}, ^{107}\text{Se}, ^{107}\text{Br}, ^{107}\text{Kr}, ^{107}\text{Rb}, ^{107}\text{Sr}, ^{107}\text{Y}, ^{107}\text{Zr}, ^{107}\text{Nb}, ^{107}\text{Mo}, ^{107}\text{Tc}, ^{107}\text{Ru}, ^{107}\text{Rh}, ^{107}\text{Pd}, ^{107}\text{Ag}, ^{107}\text{Cd}, ^{107}\text{In}, ^{107}\text{Sn}, ^{107}\text{Sb}, ^{107}\text{Te}, ^{107}\text{I}, ^{107}\text{Xe}, ^{107}\text{Cs}$	5.181098×10^{-56}	$[4.24061 \times 10^{-56}, 6.07269 \times 10^{-56}]$	PX
108	24	$^{108}\text{Ge}, ^{108}\text{As}, ^{108}\text{Se}, ^{108}\text{Br}, ^{108}\text{Kr}, ^{108}\text{Rb}, ^{108}\text{Sr}, ^{108}\text{Y}, ^{108}\text{Zr}, ^{108}\text{Nb}, ^{108}\text{Mo}, ^{108}\text{Tc}, ^{108}\text{Ru}, ^{108}\text{Rh}, ^{108}\text{Pd}, ^{108}\text{Ag}, ^{108}\text{Cd}, ^{108}\text{In}, ^{108}\text{Sn}, ^{108}\text{Sb}, ^{108}\text{Te}, ^{108}\text{I}, ^{108}\text{Xe}, ^{108}\text{Cs}$	8.079738×10^{-56}	$[6.64376 \times 10^{-56}, 9.4357 \times 10^{-56}]$	PX
109	25	$^{109}\text{Ge}, ^{109}\text{As}, ^{109}\text{Se}, ^{109}\text{Br}, ^{109}\text{Kr}, ^{109}\text{Rb}, ^{109}\text{Sr}, ^{109}\text{Y}, ^{109}\text{Zr}, ^{109}\text{Nb}, ^{109}\text{Mo}, ^{109}\text{Tc}, ^{109}\text{Ru}, ^{109}\text{Rh}, ^{109}\text{Pd}, ^{109}\text{Ag}, ^{109}\text{Cd}, ^{109}\text{In}, ^{109}\text{Sn}, ^{109}\text{Sb}, ^{109}\text{Te}, ^{109}\text{I}, ^{109}\text{Xe}, ^{109}\text{Cs}, ^{109}\text{Ba}$	2.388884×10^{-56}	$[1.95968 \times 10^{-56}, 2.79501 \times 10^{-56}]$	PX
110	24	$^{110}\text{As}, ^{110}\text{Se}, ^{110}\text{Br}, ^{110}\text{Kr}, ^{110}\text{Rb}, ^{110}\text{Sr}, ^{110}\text{Y}, ^{110}\text{Zr}, ^{110}\text{Nb}, ^{110}\text{Mo}, ^{110}\text{Tc}, ^{110}\text{Ru}, ^{110}\text{Rh}, ^{110}\text{Pd}, ^{110}\text{Ag}, ^{110}\text{Cd}, ^{110}\text{In}, ^{110}\text{Sn}, ^{110}\text{Sb}, ^{110}\text{Te}, ^{110}\text{I}, ^{110}\text{Xe}, ^{110}\text{Cs}, ^{110}\text{Ba}$	6.937885×10^{-56}	$[5.69107 \times 10^{-56}, 8.11767 \times 10^{-56}]$	PX
111	25	$^{111}\text{As}, ^{111}\text{Se}, ^{111}\text{Br}, ^{111}\text{Kr}, ^{111}\text{Rb}, ^{111}\text{Sr}, ^{111}\text{Y}, ^{111}\text{Zr}, ^{111}\text{Nb}, ^{111}\text{Mo}, ^{111}\text{Tc}, ^{111}\text{Ru}, ^{111}\text{Rh}, ^{111}\text{Pd}, ^{111}\text{Ag}, ^{111}\text{Cd}, ^{111}\text{In}, ^{111}\text{Sn}, ^{111}\text{Sb}, ^{111}\text{Te}, ^{111}\text{I}, ^{111}\text{Xe}, ^{111}\text{Cs}, ^{111}\text{Ba}, ^{111}\text{La}$	2.598681×10^{-56}	$[2.13033 \times 10^{-56}, 3.04209 \times 10^{-56}]$	PX
112	25	$^{112}\text{As}, ^{112}\text{Se}, ^{112}\text{Br}, ^{112}\text{Kr}, ^{112}\text{Rb}, ^{112}\text{Sr}, ^{112}\text{Y}, ^{112}\text{Zr}, ^{112}\text{Nb}, ^{112}\text{Mo}, ^{112}\text{Tc}, ^{112}\text{Ru}, ^{112}\text{Rh}, ^{112}\text{Pd}, ^{112}\text{Ag}, ^{112}\text{Cd}, ^{112}\text{In}, ^{112}\text{Sn}, ^{112}\text{Sb}, ^{112}\text{Te}, ^{112}\text{I}, ^{112}\text{Xe}, ^{112}\text{Cs}, ^{112}\text{Ba}, ^{112}\text{La}$	7.601681×10^{-56}	$[6.25364 \times 10^{-56}, 8.87409 \times 10^{-56}]$	PX
113	24	$^{113}\text{Se}, ^{113}\text{Br}, ^{113}\text{Kr}, ^{113}\text{Rb}, ^{113}\text{Sr}, ^{113}\text{Y}, ^{113}\text{Zr}, ^{113}\text{Nb}, ^{113}\text{Mo}, ^{113}\text{Tc}, ^{113}\text{Ru}, ^{113}\text{Rh}, ^{113}\text{Pd}, ^{113}\text{Ag}, ^{113}\text{Cd}, ^{113}\text{In}, ^{113}\text{Sn}, ^{113}\text{Sb}, ^{113}\text{Te}, ^{113}\text{I}, ^{113}\text{Xe}, ^{113}\text{Cs}, ^{113}\text{Ba}, ^{113}\text{La}$	1.656442×10^{-56}	$[1.35994 \times 10^{-56}, 1.93681 \times 10^{-56}]$	PX
114	25	$^{114}\text{Se}, ^{114}\text{Br}, ^{114}\text{Kr}, ^{114}\text{Rb}, ^{114}\text{Sr}, ^{114}\text{Y}, ^{114}\text{Zr}, ^{114}\text{Nb}, ^{114}\text{Mo}, ^{114}\text{Tc}, ^{114}\text{Ru}, ^{114}\text{Rh}, ^{114}\text{Pd}, ^{114}\text{Ag}, ^{114}\text{Cd}, ^{114}\text{In}, ^{114}\text{Sn}, ^{114}\text{Sb}, ^{114}\text{Te}, ^{114}\text{I}, ^{114}\text{Xe}, ^{114}\text{Cs}, ^{114}\text{Ba}, ^{114}\text{La}, ^{114}\text{Ce}$	8.683578×10^{-56}	$[7.13025 \times 10^{-56}, 1.01521 \times 10^{-55}]$	PX
115	25	$^{115}\text{Se}, ^{115}\text{Br}, ^{115}\text{Kr}, ^{115}\text{Rb}, ^{115}\text{Sr}, ^{115}\text{Y}, ^{115}\text{Zr}, ^{115}\text{Nb}, ^{115}\text{Mo}, ^{115}\text{Tc}, ^{115}\text{Ru}, ^{115}\text{Rh}, ^{115}\text{Pd}, ^{115}\text{Ag}, ^{115}\text{Cd}, ^{115}\text{In}, ^{115}\text{Sn}, ^{115}\text{Sb}, ^{115}\text{Te}, ^{115}\text{I}, ^{115}\text{Xe}, ^{115}\text{Cs}, ^{115}\text{Ba}, ^{115}\text{La}, ^{115}\text{Ce}$	3.337931×10^{-56}	$[2.74064 \times 10^{-56}, 3.90266 \times 10^{-56}]$	PX
116	26	$^{116}\text{Se}, ^{116}\text{Br}, ^{116}\text{Kr}, ^{116}\text{Rb}, ^{116}\text{Sr}, ^{116}\text{Y}, ^{116}\text{Zr}, ^{116}\text{Nb}, ^{116}\text{Mo}, ^{116}\text{Tc}, ^{116}\text{Ru}, ^{116}\text{Rh}, ^{116}\text{Pd}, ^{116}\text{Ag}, ^{116}\text{Cd}, ^{116}\text{In}, ^{116}\text{Sn}, ^{116}\text{Sb}, ^{116}\text{Te}, ^{116}\text{I}, ^{116}\text{Xe}, ^{116}\text{Cs}, ^{116}\text{Ba}, ^{116}\text{La}, ^{116}\text{Ce}, ^{116}\text{Pr}$	2.221062×10^{-56}	$[1.81883 \times 10^{-56}, 2.60226 \times 10^{-56}]$	PX

Continued on next page

A	N	Represented species / annotations	Y_A at $\eta_{10} = 6.094$	Six-anchor range	Class
117	25	$^{117}\text{Br}, ^{117}\text{Kr}, ^{117}\text{Rb}, ^{117}\text{Sr}, ^{117}\text{Y}, ^{117}\text{Zr}, ^{117}\text{Nb},$ $^{117}\text{Mo}, ^{117}\text{Tc}, ^{117}\text{Ru}, ^{117}\text{Rh}, ^{117}\text{Pd}, ^{117}\text{Ag},$ $^{117}\text{Cd}, ^{117}\text{In}, ^{117}\text{Sn}, ^{117}\text{Sb}, ^{117}\text{Te}, ^{117}\text{I}, ^{117}\text{Xe},$ $^{117}\text{Cs}, ^{117}\text{Ba}, ^{117}\text{La}, ^{117}\text{Ce}, ^{117}\text{Pr}$	5.209952×10^{-56}	$[4.2719 \times 10^{-56}, 6.09784 \times 10^{-56}]$	PX
118	25	$^{118}\text{Br}, ^{118}\text{Kr}, ^{118}\text{Rb}, ^{118}\text{Sr}, ^{118}\text{Y}, ^{118}\text{Zr}, ^{118}\text{Nb},$ $^{118}\text{Mo}, ^{118}\text{Tc}, ^{118}\text{Ru}, ^{118}\text{Rh}, ^{118}\text{Pd}, ^{118}\text{Ag},$ $^{118}\text{Cd}, ^{118}\text{In}, ^{118}\text{Sn}, ^{118}\text{Sb}, ^{118}\text{Te}, ^{118}\text{I}, ^{118}\text{Xe},$ $^{118}\text{Cs}, ^{118}\text{Ba}, ^{118}\text{La}, ^{118}\text{Ce}, ^{118}\text{Pr}$	7.839828×10^{-56}	$[6.45036 \times 10^{-56}, 9.15121 \times 10^{-56}]$	PX
119	26	$^{119}\text{Br}, ^{119}\text{Kr}, ^{119}\text{Rb}, ^{119}\text{Sr}, ^{119}\text{Y}, ^{119}\text{Zr}, ^{119}\text{Nb},$ $^{119}\text{Mo}, ^{119}\text{Tc}, ^{119}\text{Ru}, ^{119}\text{Rh}, ^{119}\text{Pd}, ^{119}\text{Ag},$ $^{119}\text{Cd}, ^{119}\text{In}, ^{119}\text{Sn}, ^{119}\text{Sb}, ^{119}\text{Te}, ^{119}\text{I}, ^{119}\text{Xe},$ $^{119}\text{Cs}, ^{119}\text{Ba}, ^{119}\text{La}, ^{119}\text{Ce}, ^{119}\text{Pr}, ^{119}\text{Nd}$	2.883556×10^{-56}	$[2.36803 \times 10^{-56}, 3.37091 \times 10^{-56}]$	PX

10 Interpretation and Claim Boundary

The strongest result is a structural and numerical transport statement within a deliberately restricted operator. Four facts are established together:

- the frozen campaign has a 59-species native PRIMAT solution through $A = 23$;
- fourteen rate-library boundary captures connect the native species set to an external graph without an inserted external abundance seed or floor;
- the checkpoint-based $64 \rightarrow 128 \rightarrow 256$ qualification supports the 256-step primary discretization at the eight preregistered masses; and
- the full six-anchor 256-step campaign has no missing or nonpositive mass-sector sum from $A = 24$ through 119, passes the full-vector source controls, and passes the separate absolute/normalized reconstruction check at the eight checkpoint masses.

The calculation does *not* establish native PRIMAT heavy-network evolution beyond $A = 23$. More strongly, the external result is not the final state of a self-consistent coupled reaction network: the native source trajectories and light bath are prescribed and are not depleted by the external transport. The external Y_A values are therefore diagnostic transport populations conditioned on that source, not conserved additions to the native PRIMAT abundance inventory.

The reaction operator is also intentionally incomplete. It retains only the frozen forward capture topology and beta-minus topology described above. Reverse photodisintegration, other capture or exchange topologies, feedback on the thermodynamic history, and competition from omitted reaction classes are not modeled. Consequently, *gap-free* means that the selected external graph has at least one represented coordinate and a positive modeled mass-sector sum for every integer $A = 24$ –119. It does not mean that a complete physical heavy-element nucleosynthesis network has been solved.

The amplitude claim is weaker than the reachability claim. At $A = 119$, the six-anchor full range is about 34.8% of the central transport population, and the largest corresponding range in the external register occurs at $A = 104$, about 35.4%. The source-tail qualification reported for the $A = 160$ –338 frontier is not asserted as an amplitude qualification of the $A = 24$ –119 register. The paper therefore reports the deep-tail populations but does not promote them to observationally significant primordial yields or source-robust abundance predictions.

11 Post- $A = 119$ Qualification and Structural Frontier

The complete $A = 1$ –119 register and its quantitative claim remain governed by the validation described above. The post- $A = 119$ calculations are independent downstream qualifications of the same frozen PRIMAT-boundary reaction rule and do not alter the authority distinction in Equation (4): PRIMAT is the native authority through $A = 23$, while all heavier coordinates are reduced-graph post-processing.

11.1 Continuum qualification over $A = 120$ –160

Every mass sector from $A = 120$ through $A = 160$ is tested at all six frozen η_{10} anchors with independent primary and verification ladders at 128, 256, and 512 substeps. The preregistered gates require primary 256-to-512 relative change no larger than 0.5%, primary 512-to-Richardson residual no larger than 0.25%, independent continuum cross-scheme difference no larger than 0.5%, primary effective order between 1.5 and 2.5, backward-Euler effective order between 0.5 and 1.5, backward-Euler 256-to-512 change no larger than 4%, together with the full-region reachability, positivity, reconstruction, zero-source, and source-linearity controls.

All six anchors pass. The global maxima and observed order ranges are

Quantity	Observed	Frozen gate
Primary 256-to-512 maximum relative change	0.232713%	$\leq 0.5\%$
Primary effective-order range	1.953864–1.954513	1.5–2.5
Backward-Euler 256-to-512 maximum relative change	1.344222%	$\leq 4\%$
Backward-Euler effective-order range	0.993655–0.993714	0.5–1.5
Continuum cross-scheme maximum relative difference	0.039950%	$\leq 0.5\%$
Primary 512-to-Richardson maximum relative residual	0.081158%	$\leq 0.25\%$

Table 7: Post- $A = 119$ continuum qualification over $A = 120$ –160. Percentages are the exact production fractions multiplied by 100 and rounded for presentation.

At every anchor the worst primary refinement sector is $A = 123$, and that sector remains inside the frozen gates. Within the declared reduced-graph operator, the calculation therefore qualifies 512-substep production over the complete $A = 120$ –160 region. This is a numerical transport statement, not native PRIMAT heavy-network evolution or an observational abundance claim.

11.2 Uncapped structural frontier and the $A = 338 \rightarrow 339$ connectivity analysis

An uncapped structural census removes the artificial maximum-mass boundary while preserving the frozen capture/beta-minus topology, exact-zero external initialization, and prohibition on imported heavy-sector numerical states. It finds no missing mass sector between $A = 24$ and the reachable maximum and locates the natural reachable frontier at $A = 338$, while the eligible frozen-rule rate library contains $A = 339$ coordinates.

A non-numerical structural-frontier connectivity analysis uses the same native authority but no heavy endpoint vector and tests the eligible $A = 339$ sector against the reachable component ending at

$A = 338$. The resulting counts are

$$N_{A=338, \text{reachable}} = 3, \quad (50)$$

$$N_{A=339, \text{eligible}} = 9, \quad (51)$$

$$N_{\text{incoming} \rightarrow A=339} = 8, \quad (52)$$

$$N_{\text{incoming from reachable}} = 0, \quad (53)$$

$$N_{A=338 \text{ outgoing to unreachable}} = 0. \quad (54)$$

Thus $A = 338$ is the natural structural reachability frontier of this *selected frozen reduced graph*: the stop is not an imposed A_{max} , and the eligible $A = 339$ coordinates are disconnected because none of their allowed incoming sources belongs to the reachable component. This statement is not a universal physical upper mass limit for nuclei and is not a complete heavy-nucleosynthesis endpoint; it is a topology result for the declared PRIMAT-anchored operator and selected ReacLib reaction classes.

11.3 Frontier numerical qualification at the 2048-step level

The uncapped graph is evaluated with 512/1024/2048 primary and backward-Euler ladders at pre-registered checkpoints approaching $A = 338$. The qualification uses the same convergence tolerances, exact-zero external initialization, prohibition on artificial abundance seeds or floors, and structural frontier rule. All six PRIMAT anchors pass independently.

The global metrics and observed order ranges are

Quantity	Observed	Frozen gate
Primary 1024-to-2048 maximum relative change	0.152089%	$\leq 0.5\%$
Primary effective-order range	1.994413–1.994464	1.5–2.5
Backward-Euler 1024-to-2048 maximum relative change	3.030155%	$\leq 4\%$
Backward-Euler effective-order range	0.974810–0.975027	0.5–1.5
Continuum cross-scheme maximum relative difference	0.172076%	$\leq 0.5\%$
Primary 2048-to-Richardson maximum relative residual	0.051037%	$\leq 0.25\%$

Table 8: Frontier 512 $\rightarrow$ 1024 $\rightarrow$ 2048 numerical qualification approaching the natural $A = 338$ structural frontier. All six anchors pass the frozen gates.

At $A = 338$, the 2048-step primary population spans $1.8474617 \times 10^{-75}$ to $5.2422539 \times 10^{-75}$ across the six anchors, and $A = 338$ is the worst checkpoint at every anchor. The primary order is essentially second order and the backward-Euler order remains near first order. The accepted claim is checkpoint-based continuum qualification approaching $A = 338$, together with full 2048-step positive support over $A = 160$ –338 in the selected reduced graph. It is not native PRIMAT heavy-network evolution, not an all-sector convergence proof over every mass from 160 through 338, and not a primordial heavy-element yield prediction.

12 Source-Tail Robustness

The late-source qualification uses the same six PRIMAT anchors, boundary-source rule, ReacLib selection, natural $A = 338$ frontier, transport equations, and 2048-step baseline resolution as the qualified frontier calculation. It retains cumulative boundary-source fractions

$$f_{\text{src}} \in \{0.99, 0.999, 0.9999\}, \quad (55)$$

with the boundary node adjusted so that the retained integral is exact under nodal trapezoidal quadrature. Source nodes after the truncation point are set to zero while the transport and loss coefficients remain active. Each truncated source is also renormalized to unit integral for the timing-shape gate, while the unrenormalized endpoint ratio is retained separately to test whether the removed late source materially contributes to the frontier amplitude. The 99% retained case is a stress diagnostic; the hard timing-shape limits are 5% maximum relative sensitivity at 99.9% retained source and 1% at 99.99% retained source, including the $A = 338$ checkpoint.

All six 2048-step baselines reproduce the qualified frontier $A = 338$ values with relative error 0.0 at stored numerical precision. All 18 rows satisfy their applicable preregistered requirements; all 12 hard-gated 99.9% and 99.99% cases pass. The fraction-level maxima are

Retained source	Cases	Max. normalized checkpoint sensitivity	Max. $ R_{338}^{\text{raw}} - 1 $
99%	6/6 PASS	1.0000%	$2.3592239273 \times 10^{-13}$
99.9%	6/6 PASS	0.1000%	$8.8373752760 \times 10^{-14}$
99.99%	6/6 PASS	0.0100%	$9.1482377229 \times 10^{-14}$

Table 9: Source-tail robustness at the six PRIMAT authority anchors. The 99% cases are stress diagnostics; the 99.9% and 99.99% cases carry the hard timing-shape gates.

Define the unrenormalized frontier ratio by

$$R_{338}^{\text{raw}}(f_{\text{src}}) = \frac{Y_{338}^{\text{truncated,raw}}}{Y_{338}^{\text{baseline}}}. \quad (56)$$

Across all anchors and all three retained-source fractions,

$$\max_{\eta_{10}, f_{\text{src}}} |R_{338}^{\text{raw}} - 1| = 2.3592239273 \times 10^{-13}. \quad (57)$$

The near-exact 1%, 0.1%, and 0.01% normalized sensitivities track the unit-integral renormalization and are therefore treated as timing-shape controls rather than independent absolute-amplitude predictions. The unrenormalized ratio in Equation (57) is the direct source-tail result: no resolved relative change larger than 2.36×10^{-13} is found in this calculation, and within the prescribed-PRIMAT-bath reduced transport the qualified $A = 338$ endpoint is insensitive to removal of the faint late boundary-source tail. Single-edge boundary-source ablations are outside this tail decision and are not used to qualify the source-tail claim.

13 Conclusion

This study establishes a PRIMAT-primary computational basis for the $A = 1$ –119 companion. In the exact frozen PRIMAT large-network configuration, the native calculation contains 59 nuclear

species through $A = 23$. The external operator begins at $A = 24$ from fourteen rate-library boundary captures, starts every external state at exact zero, and uses no heavy-sector endpoint vector from another transport calculation, artificial external seed, or abundance floor.

A separate prospective native validation supports the locked CPTG mass-seven transport law in the primary network. Across one CPTG transported baryon-density coordinate and all six frozen PRIMAT authority anchors, the live ${}^7\text{Li}/{}^7\text{Be}$ RHS and analytic-Jacobian sink reproduces the fixed $1/\pi$ survival to a worst relative deviation of 0.018083%, while the maximum disturbances in D/H , ${}^3\text{He}/\text{H}$, and Y_p are 3.87×10^{-9} , 5.89×10^{-10} , and 3.83×10^{-12} , respectively. The result is a direct live-network test rather than a post-processing division or observational fit. The separate CPTG radiation-coordinate mapping remains outside the validated scope.

The $A = 1$ –119 numerical method is qualified with independent primary and backward-Euler asymptotic sequences at eight preregistered checkpoint masses. The primary sequence is near second order, backward Euler is near first order, the independently extrapolated checkpoint continua agree within 0.206809%, and the 256-step primary checkpoint results lie within 0.064104% of their Richardson limits. The six-anchor production calculation passes the frozen graph, full-state nonnegativity, full-vector source-linearity, all-sector reachability, and eight-checkpoint absolute/normalized reconstruction gates through $A = 119$.

Within that declared scope, the external operator has strictly positive modeled mass-sector support for every $A = 24$ –119 at all six anchors, and the complete register represents every integer mass number $A = 1$ –119 with $A = 5$ explicitly separated as a non-abundance resonance-gap annotation. The result is not promoted to native PRIMAT heavy-network evolution, a closed-network primordial heavy-element abundance prediction, or an observational yield claim. The native source and light bath remain prescribed and only selected forward reaction classes are included. The source-tail qualification at $A = 160$ –338 is not asserted as an amplitude qualification of the $A = 24$ –119 register.

The same reduced-graph rule has a natural structural frontier at $A = 338$, with the eligible $A = 339$ sector disconnected because none of its allowed incoming sources belongs to the reachable component. The qualified $512 \rightarrow 1024 \rightarrow 2048$ frontier calculation passes all six anchors: the worst primary 1024-to-2048 checkpoint change is 0.152089%, the worst continuum cross-scheme difference is 0.172076%, the worst 2048-to-Richardson residual is 0.051037%, and the observed method orders remain near second and first order. Positive 2048-step support is present throughout $A = 160$ –338. The source-tail qualification satisfies the applicable preregistered requirements in all 18 retained-source cases, including all 12 hard-gated cases, and the maximum absolute departure of the unrenormalized truncated-to-baseline $A = 338$ ratio from unity is 2.36×10^{-13} . Within the prescribed-PRIMAT-bath reduced transport, the qualified $A = 338$ frontier is therefore insensitive to removal of the faint late boundary-source tail. These post- $A = 119$ results remain checkpoint-based numerical, structural, and source-tail qualifications of the selected reduced graph; they are not native PRIMAT heavy-network evolution, an all-sector convergence proof, or a primordial heavy-element yield prediction.

A Evidence Ledger and Reproducibility

All package identities, preregistration identities, result-object identities, transcript identities, and SHA-256 digests used to audit this computational companion are consolidated in this section. Each evidence object is described separately. Where an object carries a scientific decision, the ledger states the decision and the reason for it. Source, freeze, and transcript objects that establish identity or execution provenance without constituting independent scientific tests are identified as provenance-only rather than being assigned an artificial pass/fail status.

A.1 Complete $A = 1$ –119 register

A.1.1 Controlling self-contained final evidence

Archive: CPTG_PRIMAT_A1A119_FINAL_EVIDENCE_20260818-071825_AUDITED_SELFCONTAINED_r01.zip

Evidence: Complete six-anchor authority package containing the native trajectory/meta pairs, frozen source and graph rules, graph register, $A = 1$ –119 register, numerical qualification, production and verification solvers, and final audit.

Decision: **PASS.** The checkpoint refinement gates are satisfied, and all six 256-step production anchors pass graph identity, finite/nonnegative-state, exact-zero-source, source-linearity, reconstruction, and strictly positive $A = 24$ –119 support. The archive manifest verifies 62/62 payload entries.

SHA-256: 39549f1eba0201999aa0953d9ed8b36d4526c5df3ef70c0f11cdc9ae57449620

A.1.2 Native-trajectory construction source

Archive: CPTG_PRIMAT_NativeBoundaryFlux_A1A119_FinalValidation_20260817_r02.zip

Evidence: Source and configuration package defining how the six PRIMAT native trajectories and boundary-flux inputs were constructed.

Decision role: **provenance-only.** This object defines the construction contract. It does not carry an independent scientific pass/fail decision.

SHA-256: 00e9cf78acae19700489400a52cac12b4264f43c96bcdcd1d65c637e5bbcb446b

A.1.3 Asymptotic numerical-qualification source and preregistration

Archive: CPTG_PRIMAT_NativeBoundaryFlux_AsymptoticContinuumQualification_20260818_r05.zip

Evidence: Executable source and frozen numerical gates for the independent primary/backward-Euler checkpoint qualification on the $64 \rightarrow 128 \rightarrow 256$ refinement sequence.

Decision role: **provenance-only.** This object fixes the numerical test before execution. The accepted numerical decision is carried by the result object below.

SHA-256: 151816ea4b34a61da53064671bd561216490b99cff4717389e268deb45390a0f

A.1.4 Native trajectory freeze object

Object: Six trajectory/meta pairs at the anchors in Equation (1).

Evidence: Identity lock for the exact native PRIMAT trajectories used by the external transport.

Decision role: **provenance-only.** This object prevents trajectory substitution. It does not itself carry a numerical pass/fail decision.

SHA-256: 97fd7b4f4f4e7466be9e45be264e62faade5b55e80d415201ed4390405080f9a

A.1.5 Accepted asymptotic-continuum result object

Object: Machine-readable two-scheme checkpoint convergence result used to qualify the 256-step production resolution.

Evidence: The result records the primary and backward-Euler refinement sequences, the independently extrapolated continuum comparison, and the Richardson residual at the preregistered checkpoint masses.

Decision: **PASS.** The worst primary refinement is 0.190056%, backward Euler is 3.176411%, continuum cross-scheme disagreement is 0.206809%, and the 256-to-Richardson residual is 0.064104%. All are inside their frozen gates, and both observed orders remain in their required bands.

SHA-256: 7bac3fc8f8b1f237314f92d2e8b522788a288429d3395e018640f09a21c45280

A.1.6 External graph register object

Object: Frozen identity of the external reduced graph used to generate the reported mass-sector continuation.

Evidence: The register fixes the graph membership used by the external transport and therefore identifies the exact reduced network to which the reported reachability claim applies.

Decision role: provenance-only. This object locks graph membership. It is not an independent pass/fail test.

SHA-256: dc2bdbbe1ad58a1a488872eea0c8947c15ca3688d006865a5fe3acae2c360a29

Archive note. An archival-export omission affecting six Windows launchers was repaired by byte-for-byte restoration from the released validation distribution. The restored files matched the hashes already recorded in the retained distribution manifest, and no scientific source, configuration, trajectory, graph, result, register, or paper-data object changed.

A.2 PRIMAT-native mass-seven live transport

A.2.1 Self-contained final evidence

Archive: CPTG_PRIMAT_NativeA7_LiveTransport_CoreTheoryValidation_FINAL_EVIDENCE_20260818.zip

Evidence: Complete native PRIMAT live-channel validation of the locked $\tau_7 = \ln \pi$ transport law across one CPTG transported coordinate and all six frozen PRIMAT anchors.

Decision: PASS. All seven paired coordinates pass. The worst survival disagreement from $1/\pi$ is 0.018083%, the independent-grid optical-depth reconstruction passes, and the non-mass-seven abundance disturbances remain negligible within the declared controls. The archive manifest verifies 18/18 payload entries.

SHA-256: 2a1f52eea6020753b08f838be54cb25005fe79506f2d05245e6dbd8bc8d0d315

A.2.2 Execution source package

Archive: CPTG_PRIMAT_NativeA7_LiveTransport_CoreTheoryValidation_20260818_r01.zip

Evidence: Executable package implementing the preregistered mass-seven gate directly in a copied PRIMAT v0.3.2 large-network source tree.

Decision role: provenance-only. The package records the executable implementation. The scientific PASS is carried by the self-contained final evidence above.

SHA-256: 384fdc86e7899a9b99d4e54883969e596d8073bead383ca5d5024d32b2f909a9

A.2.3 Frozen preregistration

Object: Mass-seven hypotheses, numerical gates, and execution contract fixed before the native paired runs.

Evidence: The preregistration records the intended test conditions before the live-channel calculation is opened.

Decision role: provenance-only. It establishes pre-execution conditions rather than a scientific result.

SHA-256: 444879ac5bbcb280f66fe99606a5da22fe99d0cf3f341c0d8cc771beaea6f67b4

A.2.4 Production transcript

Object: Complete fourteen-run gate-off/gate-on execution record for the seven tested coordinates.

Evidence: The transcript preserves the production execution sequence and outputs used by the final evidence package.

Decision role: execution provenance. It supports the final PASS but is not a separate scientific pass/fail object.

SHA-256: db24518c9947c0a9a167fe9c4fe18300cd84849c3054e67067a1dcb9f925f0a2

A.2.5 Independent final-audit transcript

Object: Independent audit record checking the completed mass-seven evidence, warnings, identities, and claim boundary.

Evidence: The audit verifies that the completed evidence is internally consistent and that the declared numerical warnings and scope limitations are preserved rather than suppressed.

Decision role: audit provenance. It supports the accepted PASS. It does not define a separate scientific test.

SHA-256: 018c79ba2be8a2caba587b7224b0842443f56b73e9fd950c0d62987d92bd8dc8

Numerical-certification note. The final evidence retains all seven adaptive-quadrature warnings and the independent-grid normalization control. The warnings are treated as numerical-certification warnings rather than failures because the native solves complete and the independent-grid optical-depth reconstruction satisfies its preregistered tolerance. The explicit validation boundary records that the separate CPTG radiation-coordinate mapping is not implemented in PRIMAT.

A.3 Post- $A = 119$ numerical and structural evidence

A.3.1 Uncapped structural census

Archive: CPTG_PRIMAT_PostA119_UncappedStructuralFrontierCensus_20260818_r08.zip

Evidence: Removes an imposed maximum-mass boundary and enumerates structural reachability under the frozen reduced-graph topology.

Decision: structural PASS. The graph remains gap-free from $A = 24$ through its reachable maximum and terminates naturally at $A = 338$, rather than at an imposed $A_{\max}$.

SHA-256: 75c4b923a1fe86be0341d788f4133cd65973d9505484419364b670c57add5445

A.3.2 $A = 120$ – 160 continuum source, code-only writer repair

Archive: CPTG_PRIMAT_PostA119_A120A160_FinerContinuumQualification_20260818_r09_HOTFIX_r01.zip

Evidence: Six-anchor 128/256/512 two-scheme qualification over every mass sector from $A = 120$ through 160. The repair changes only output writing and leaves the scientific gates unchanged.

Decision: PASS. The observed maxima are 0.232713% for primary refinement, 1.344222% for backward-Euler refinement, 0.039950% for continuum cross-scheme disagreement, and 0.081158% for the Richardson residual. Each is inside its frozen limit, and both method orders remain inside their required bands.

SHA-256: bb6256de6e05fa47cfec33387444a0a204fb180f0d26688fb4fd53a832cec2ac

A.3.3 $A = 338 \rightarrow 339$ structural-frontier connectivity analysis

Archive: CPTG_PRIMAT_PostA119_A338A339_StructuralFrontierForensic_20260818_r10.zip

Evidence: Non-numerical connectivity audit of the eligible $A = 339$ rate-library sector against the reachable component ending at $A = 338$.

Decision: structural PASS. The audit finds three reachable $A = 338$ coordinates, nine eligible $A = 339$ coordinates, eight incoming $A = 339$ edges, and zero incoming edges whose sources are reachable. The $A = 339$ sector is therefore a disconnected rate island under the selected rule.

Recorded classification: A339_DISCONNECTED_RATE_ISLAND__ALL_INCOMING_SOURCES_UNREACHABLE

SHA-256: 87d32d3e002b83d405ec5b8043ed5be1934f1027fdfce218ec403e4a4dc27a6f

A.3.4 128/256/512 frontier-approach source package

Archive: CPTG_PRIMAT_PostA119_A160A338_FrontierApproachContinuumQualification_20260818_r11.zip

Evidence: Six-anchor two-scheme convergence test approaching the $A = 338$ frontier at a 512-step ceiling.

Decision: fail-closed NOT_QUALIFIED. The primary 256-to-512 change is 2.387611%, the backward-Euler change is 11.507298%, the continuum cross-scheme difference is 2.324233%, and the 512-to-Richardson residual is 0.879535%. These exceed their unchanged 0.5%, 4%, 0.5%, and 0.25% gates, respectively. The observed method orders remain regular, so the result is classified as insufficient numerical resolution rather than a loss of the $A = 338$ structural frontier.

SHA-256: cae77e1c79b16dc81ae0ceac3f63c117cb6c881bfa629e6ae61f63b717a390f9

A.3.5 512/1024/2048 frontier source and preregistration

Archive: CPTG_PRIMAT_PostA119_A160A338_FinerFrontierContinuumQualification_20260818_r12.zip

Evidence: Finer six-anchor primary/backward-Euler checkpoint qualification approaching $A = 338$, using the same scientific convergence limits at 512/1024/2048 resolution.

Decision: PASS. The observed maxima are 0.152089% for primary refinement, 3.030155% for backward-Euler refinement, 0.172076% for continuum cross-scheme disagreement, and 0.051037% for the 2048-to-Richardson residual. Each is inside the unchanged gate. The method orders remain inside their required bands, and the 2048-step solution has positive support over $A = 160$ –338.

SHA-256: d73b2a14626c0217784a8d3fb888b2b4a59b858daa71f7ef276090f1deda0366

A.3.6 Audited frontier final evidence

Archive: CPTG_PRIMAT_PostA119_A160A338_FinerFrontierContinuumQualification_FINAL_EVIDENCE_20260818_r01.zip

Evidence: Self-contained final authority for the accepted 512/1024/2048 frontier calculation, including the six-anchor transcript, machine-readable metrics, preregistration, decision, and claim boundary.

Decision: PASS. The archive preserves the accepted six-anchor finer-frontier result in which every frozen convergence gate is satisfied. It is the numerical authority for the frontier claim; the archived 512-step non-qualification remains in the ledger as execution provenance.

SHA-256: b0733da182a4a31a810ccd97e23b33389e11099b5b8bfc72e32501157e7dd68c

A.3.7 Windows execution transcript for the $A = 120$ –160 qualification and structural-frontier connectivity analysis

Archive: POST_A119_WINDOWS_EXECUTION_TRANSCRIPT_20260818.txt

Evidence: Console execution provenance for the accepted $A = 120$ –160 numerical qualification and the $A = 338 \rightarrow 339$ structural connectivity analysis.

Decision role: provenance-only. The transcript records execution. The PASS decisions are carried by their corresponding result and evidence objects above.

SHA-256: 585d5b44fdb808db731a900733c530f3f52aa415a34daf8a91b04beee0f514e2

A.4 Source-tail evidence

A.4.1 Frozen source-tail preregistration

Object: Identity lock for the retained-source fractions, truncation construction, timing-shape gates, and tail claim boundary.

Evidence: The preregistration fixes the source-tail test before production, including the 99%, 99.9%, and 99.99% retained-source cases and their declared gate roles.

Decision role: provenance-only. It establishes the pre-execution test contract and does not itself carry a PASS/FAIL result.

SHA-256: acc25bffbe7b18ed886a162fee67b9a61571bb4a6a57fc0db3bbe6c73f660f6a

A.4.2 Controlling tail-only self-sealed final evidence

Archive: CPTG_PRIMAT_PostA119_A160A338_SourceTail_FINAL_SELFSEALED_EVIDENCE_20260819_r01.zip

Evidence: Self-contained six-anchor tail-only rerun with six 2048-step baselines and eighteen retained-source rows at 99%, 99.9%, and 99.99% cumulative source. The archive internally preserves the frozen source-tail preregistration, runtime-binding provenance, scientific source, execution transcript, and result objects needed to reconstruct the reported tail test.

Decision: PASS. All 18 rows satisfy their applicable preregistered requirements, and all 12 hard-gated rows pass. The maximum unrenormalized $A = 338$ departure from baseline is $2.3592239273284576 \times 10^{-13}$. The separate leave-one-out branch is excluded from this decision. The archive manifest verifies 35/35 payload entries.

SHA-256: d78aae075d3fe579874d3002fdee6b4d47d98712df9dd1ff85d2f9022e4f246a

Reconstruction environment

For software reconstruction of the external transport and final validation, the isolated transport environment is pinned to NumPy 2.4.6, Numba 0.66.0, and `pynucastro` 2.11.0. The PRIMAT extraction contract requires PRIMAT v0.3.2. The authority-adoption objects record zero historical heavy-sector numerical inputs, no imported endpoint vectors from other transport calculations, and no regeneration of the frozen native trajectories.

Evidence and code availability

The manuscript, evidence archives, and supporting code are intended for public release and can be located at: <https://github.com/CLG2025/CPTG/tree/main/nuclear-reactions>

References

- [1] Carter L. Glass Jr., *A Universal Geometric Theory of Nuclear Reactions in CPTG: The Four-Sector Foundation of Vertex, Bridge, Closure, and Saturation*, 2026, DOI: 10.5281/zenodo.22439971.
- [2] Carter L. Glass Jr., *Geometric Nuclear Reaction Theory in CPTG: Deuterium-Proton Capture and Primordial Mass-Seven Transport*, 2026, DOI: 10.5281/zenodo.22425309.
- [3] C. Pitrou, A. Coc, J.-P. Uzan, and E. Vangioni, “Precision big bang nucleosynthesis with improved Helium-4 predictions,” *Physics Reports* **754**, 1–66 (2018), DOI: 10.1016/j.physrep.2018.04.005.
- [4] R. H. Cyburt et al., “The JINA REACLIB Database: Its Recent Updates and Impact on Type-I X-Ray Bursts,” *The Astrophysical Journal Supplement Series* **189**, 240–252 (2010), DOI: 10.1088/0067-0049/189/1/240.
- [5] A. I. Smith et al., “pynucastro: A Python Library for Nuclear Astrophysics,” *The Astrophysical Journal* **947**, 65 (2023), DOI: 10.3847/1538-4357/acbaff.
- [6] D. R. Tilley, C. M. Cheves, J. L. Godwin, G. M. Hale, H. M. Hofmann, J. H. Kelley, C. G. Sheu, and H. R. Weller, “Energy Levels of Light Nuclei $A = 5, 6, 7$,” *Nuclear Physics A* **708**, 3–163 (2002), DOI: 10.1016/S0375-9474(02)00597-3.
- [7] CPTG, *Supporting Python Models, Benchmark Implementations, and Research References for Curvature Polarization Transport Gravity*, available at <https://github.com/CLG2025/CPTG>.